



FINC-4900 International Finance
Student Book of Handouts
Fall 2026



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International Paper-Money Systems, 1971– ?

1. THE EMERGENCE OF PAPER-MONEY STANDARDS

The Fed's suspension of payments in August 1971 created in one mighty stroke a great number of paper monies. Before that date, all national currencies were basically fractional-reserve certificates for gold (via the U.S. dollar). The suspension "transubstantiated" these certificates into paper monies, with all the concomitant effects we have discussed above.

Many observers believed that the world would remain so fragmented. Advocates of paper money thought this was all well and good, because each government was now at last autonomous in its monetary policy. Others looked with horror on the reality of fluctuating exchange rates, which undermined the international division of labor and thus created misery and death for many millions of people. But the world did not long remain in monetary fragmentation. The events of the past thirty-five years illustrate that there is a tendency for the spontaneous emergence of international paper-money standards. Today the reasons for this development are not difficult to discern.¹

¹However, they have been overlooked for many years. The standard explanation of this phenomenon is indeed untenable. In this account,

One driving force of this process was of course the presence of private individuals and firms operating in many countries. These persons and organizations constantly look for ways of saving money, for example, by minimizing the costs of holding money. One way to do this is to make the bulk of one's payments in terms of only one kind of money. But this driving force, formidable though it might appear in our present time of multinational corporations, does not go a very long way in explaining the *emergence* of an international paper-money standard. The reason is that multinational corporations do not play a great role in a world of wildly fluctuating exchange rates. They operate profitably and grow to significant size only when the political framework has already stabilized the foreign exchanges. That is, by and large they come into play only once a monetary standard already exists.

This brings us to the main driving force of the emergence of international paper-money monetary standards, namely, the constant appetite of governments for additional revenue. Most governments that obtain income mostly from their own citizens—be it in the form of taxation or in the form of debt—have a rather small revenue base. To increase revenues they have by and large only two strategies: (1) induce foreign citizens to buy its bonds; (2) adopt policies that make their own citizens richer, so that they can pay more taxes and buy more government bonds.

No investor intentionally wastes his money. When he buys the bonds of a foreign government, he seeks to earn interest. He would abstain from the deal altogether if he had good reasons to believe that the money would be wasted. If he must fear, for example, that the debtor-government will simply print the money needed to pay back the credit, thus

the different countries of the post-1971 world economy are compared to the participants of a barter economy. The point of an international paper-money standard is then to allow for a greater volume of exchanges as compared with the initial barter situation. But these hypothetical additional exchanges are conceivable only if international money already exists—and the point is to explain how this happens in the first place.

provoking a fall in the exchange rate, he will not buy its bonds at all.² Thus the question is what a susceptible debtor-government can do to dispel such fears. The answer is that it must establish institutional safeguards against a falling exchange rate of its currency in terms of the currency used by its creditors.

The same considerations come into play if we turn to the second fundamental strategy for increasing public revenue. The idea is very simple: adopt policies that permit the citizens to make themselves richer so that they can pay more taxes and buy more government bonds. But the crucial point is that the productive capacity of a nation entirely depends on the capital stock it can use. This capital stock could be increased through savings from current income. But the accumulation of capital through savings can take many years and decades until it reaches any significant proportion. And during this time the government must keep the tax load as small as possible. Unfortunately such restraint requires more virtue than most governments have. The only remaining way out is, again, to encourage foreigners to provide capital that they have accumulated in their home countries—in other words, to make “foreign direct investments” in that country. But this reverts back to our previous consideration. In a paper-money world, foreign capital can be attracted only under sufficient institutional safeguards.

Four such institutions have played a significant role in the past thirty years. They go a long way in explaining the emergence of international paper money standards.

First, debtor-governments have floated bonds that were denominated in a foreign paper money, the production of which they cannot directly control; preferably this would be the paper money used in the country of its creditors. In the

²Many observers have been deluded about this point because in the period between 1948 and 1989 huge amounts of western credit have been given to corrupt governments in the Third World without any financial safeguards whatever. But these were political loans. Their purpose was not to earn monetary interest, but to buy the support of these governments during the Cold War.

past twenty years, this has become a widespread practice. Today many governments issue bonds that are denominated in U.S. dollars or euros.

Second, the government and/or the monetary authority of the country in which the creditors reside could give explicit or implicit guarantees to maintain the market exchange rate. It is widely assumed, for example, that the U.S. Federal Reserve gave such guarantees in the 1990s to the governments of Mexico, Singapore, Malaysia, Thailand, and other countries of the Far East. The great disadvantage of this practice is that it entails moral hazard for the beneficiaries. The receiving governments can set out to inflate their currencies without fearing any negative repercussions on the exchange rate. And thus they are able to expropriate not only their own population, but also the population of the country in which its creditors reside.

In the above-mentioned cases, the exchange-rate policies of the Fed had the effect of making U.S. citizens pay for the monetary abuses of the governments of Mexico and other countries. (They pay by constantly delivering goods and services to Mexico that they could have enjoyed themselves and in payment for which nothing but peso-denominated paper slips are sent to the U.S.) Because no diplomatic solution could be found for this problem, the Fed eventually abolished its policy and thus provoked financial crises in Mexico (1994) and various other countries, especially in Asia (1997). Since then, there have been no new major experiments with exchange-rate stabilization.

Third, debtor-governments have set up currency boards, thus transforming their currency into a substitute for a foreign paper money. This technique too is widely used today, for example, in Hong Kong, Bulgaria, Estonia, Lithuania, Bosnia, and Brunei.

Fourth, debtor-governments have abandoned the use of the national currency altogether and adopted the use of the paper money used by the creditors. Economists call such a policy “dollarization,” even when the government adopts not the U.S. dollar, but a different foreign paper money. Among

recently dollarized countries are Ecuador, El Salvador, Kosovo, and Montenegro.³

We conclude that the driving force for the emergence of an international paper-money standard is the quest of governments for additional funds, which most of them can obtain only from abroad. And our analysis also explains which paper monies will tend to be chosen as international standards: the paper monies that are legal tender in the territories with the largest capital markets. In the period under consideration, these territories happened to be the U.S., Japan, and Europe. It is therefore not surprising that the U.S. dollar and the yen have emerged as regional monetary standards of the world economy.

The operation of the same mechanism could be observed in the case of the German mark, which during the 1990s was used as a (unofficial) parallel currency in many countries of the former East Bloc. And it currently brings about a wider geographical circulation of the euro, which was only created in early 1999, and which did not exist in the form of banknotes before the year 2002. The creation of the euro is of some interest because here an international standard did not emerge through the unilateral adoption of paper money used on foreign capital markets, but through merger. This form of monetary integration could play a role in the future development of the international monetary order. We will therefore take a brief look at it.⁴

³Dollarization is the most complete way for a government to renounce its control over the production of money. By contrast, the creation of a currency board still leaves open the possibility of a quick return to a national paper money. See Nikolay Gertchev, "The Case against Currency Boards," *Quarterly Journal of Austrian Economics* 5, no. 4 (2002).

⁴We have analyzed the development of the European Monetary System and the emergence of the euro in more detail in Jörg Guido Hülsmann, "Schöne neue Zeichengeldwelt," epilogue to Murray Rothbard, *Das Schein-Geld-System* (Gräfelfing: Resch, 2000), pp. 111–54.

2. PAPER-MONEY MERGER: THE CASE OF THE EURO

After the demise of the Bretton Woods system in 1971, the countries of Western Europe for a few years fell into monetary disarray and the fiscal anarchy that typically goes in hand with it. Each national government issued its own paper money and started piling up public debts to an unheard-of extent. The newly available funds were used to expand government welfare services. For a while, things looked rosy and only a few fiscal conservatives bemoaned the new laxity. But soon even the champions of the new policy began to understand that the new monetary order affected their interests in very tangible negative ways. Exchange rates fluctuated very widely and effectively prevented the further development of international trade. The division of labor in Europe, one of the most densely populated regions of the world, lagged behind the American economy and even—or so it seemed in those days—the Soviet economy. It was therefore but a question of time until tax revenues would fall far behind the revenues of the great competitors of the European governments: the governments of the U.S. and of the Soviet Union. Something had to be done.

The first attempts at stabilizing the exchange rates between European paper monies had failed miserably. Then, at a December 1978 conference in the German city of Bremen, the core governments of the European Economic Community, as it was called in those days, launched a new attempt at integration: the European Monetary System (EMS). The EMS was a cartel of the national paper-money producers, who agreed to coordinate their policies in order to stabilize exchange rates between their monies at certain levels or “parities.” As in the case of the previous international banking cartels, the EMS essentially relied on the self-restraint of its members. “Coordination” meant in practice that the least inflationary money producer set the pace of inflation for all others. If for example the supply of the Italian lira increased by 30 percent, whereas the supply of French francs increased only by 15 percent, it was very likely that the lira would drop on the foreign exchanges *vis-à-vis* the franc. In order to maintain the lira-franc parity, it was necessary either that the Banque de France

increase the production of francs, or that the Banco d'Italia reduce its production of lira. As we have said, the EMS essentially relied on self-restraint; thus in our example the Banca d'Italia would be expected to reduce its lira production. If for political reasons it was unwilling to do this, there would be a "realignment" of the parity, and stabilization would henceforth seek to preserve the new parity.

Now by far the least inflationary money producer happened to be the German Bundesbank. Accordingly, for the next twelve years or so, the main problem of European monetary politics was that the Bundesbank did not inflate the supply of the Deutsche mark quite enough to suit the needs of foreign governments. The latter were therefore forced to cut down their money production. It also came to frequent changes or realignments of parities. The problem was settled only in the early 1990s, when the German government sought to take over former communist East Germany and needed the consent of its major western partners. The price for that consent was the abdication of the Deutsche mark.⁵ Within a few years, the political and legal foundations were laid for merging the different national paper-money producers into one organization: the European System of Central Banks (ESCB), the coordination of which lay in the hands of the European Central Bank (ECB). The ESCB started its operations in January 1999 and three years later issued its euro notes and coins.

From an economic and ethical point of view, the euro does not bring any new aspects into play. It is just another paper money. In public debate, the introduction of the euro has often been justified by the benefits that spring from monetary integration. It is true that such benefits exist. But, as we have repeatedly emphasized in our study, these benefits can be obtained much more conveniently and assuredly by allowing the citizens to choose the best money they can get. If this had been the policy of the European governments, it would not

⁵For an insider view of the conflicts and struggles behind European monetary integration see Bernard Connolly, *The Rotten Heart of Europe* (London: Faber and Faber, 1995).

have prevented European monetary unification. But this would have been a *spontaneous* unification. Gold and silver coins would have been the harbingers of monetary integration under the auspices of liberty and responsibility.

But the European governments never intended to grant their citizens the sovereignty that they have according to the letter of written constitutions. The governments wished above all to stay in control of monetary affairs. It was out of the question to abolish the privileges for paper money. European monetary integration had to be built on paper money, for the sole reason that paper money is the source of virtually unlimited government income, at the expense of the population. This is a point that cannot be emphasized enough. The euro was not introduced out of any economic necessity. All true benefits that it conveys could have been conveyed much better through commodity monies such as gold and silver.

The story of the euro is not a success story, unless the standard of success is to be seen in the expansion of government power. Yet the euro story could be seen as a model for further monetary integration on a global scale.

3. THE DYNAMICS OF MULTIPLE PAPER-MONEY STANDARDS

The international monetary order at the outset of the twentieth-first century is characterized by the presence of several competing paper-money systems. Each of these systems is hierarchical, with standard paper money on the one hand, and a plethora of secondary and tertiary currency on the other hand. The three most important standards are the yen, the dollar, and the euro. Only these standard monies are true *monies*—paper monies or electronic monies. The secondary currencies in each of the three systems are not monies at all; rather they are national certificates for the standard money, issued on a fractional-reserve basis by a national authority (usually called a “central bank” or “currency board”). And then there are tertiary currencies that are also fractional-reserve certificates, in particular, the demand deposits of commercial banks.

We have already discussed the dialectical power relationship between national central banks and commercial banks under the gold standard. Similar considerations apply in the present case. The difference is, of course, that there is no longer any commodity-money standard that could act as a natural restraint on the drive to inflate. Even more to the point, it is at present equally impossible to restrain this drive by *legal* means, because the principle of national sovereignty still holds.⁶ As a consequence, the secondary and tertiary layers have, in the present order of things, far greater power to inflate than they ever had before. Let us explain this in more detail.

The producers of international paper money have the privilege of picking those who receive the newly printed notes first, and they have political leverage on the producers of the secondary currencies in times of crises. But this dependency is mutual. Consider that, within each nation, the commercial banks can exploit the moral hazard of the central bank. They can push inflation with the good hope that the central bank will bail them out in times of a liquidity crisis. In an international paper-money system, the same mechanism bears on the

⁶In each country the banking industry is regulated to curb some of the excesses of fractional-reserve banking. The strength of these regulations varies from one country to another, and the banks operating from the least regulated countries have therefore a competitive advantage over the other banks. Yet the risks of their enhanced activities are experienced even in the more regulated countries, because international business ties create spill-over effects. More recently, therefore, a number of governments have tried to set up international standards for the regulation of the banking industry. In particular, they seek to impose on fractional-reserve banks a minimum capital-reserve requirement on their loans; and to make this capital-reserve also dependent on the risk of each individual credit, as evaluated according to formulas developed by an international committee. The activities of the regulators are coordinated by the Bank for International Settlements (BIS) in Basel, Switzerland. They have recently published a detailed proposal known as the “Basel II Agreement” (June 2004). Let us emphasize again that, in light of our analysis, it would be more commensurate to simply abolish the legal privileges of the banking industry, rather than to layer additional international regulations atop the manifold national regulations.

relationship between the producer of the standard money and the producers of the secondary and tertiary currencies. The latter have an incentive to push inflation and speculate on bailouts.

If the producer of the standard money gives in to these demands, the exchange rate of his money will drop and the price level will increase. Both events will tend to make his money less attractive as a financial asset. Moreover, both events will tend to make the economies in which his money is used less attractive places to invest in. If he undertakes a major bailout, he even risks a hyperinflation and subsequent destruction of his product.

But our producer of standard money also runs into difficulties if he does *not* give in to any bailout demands. Consider the following scenario. The hypothetical country Ruritania has a currency board issuing Rurs backed up with dollars. The dollar exchange-rate of the Rur has been set at a very low level in order to encourage exports to the U.S. The dollars that stream into Ruritania as payment for these exports are not spent on U.S. products, but stockpiled as reserves in the vaults of the local central bank. Suppose further that the commercial banks of that country have created a huge amount of credit out of thin air (inflation) and are now in a liquidity crisis. The Ruritanian currency board turns for help to the U.S. Fed, but the Fed refuses to bail out the Ruritanian banks. At that point, the currency board could threaten to sell all its dollars for euros, thus putting the country on the euro standard. Depending on the size of Ruritania, this action would have a more or less notable impact on the dollar-euro exchange rate. It would harm the U.S. capital markets and thus provide an incentive for investors to leave Manhattan and Chicago, and to turn to Frankfurt and Paris. Moreover, if we assume that Ruritania is a very large country with substantial dollar reserves even by world standards, then the mere announcement that the Ruritanian government will switch to the euro standard might incite other member countries of the dollar standard to do the same. This could precipitate the dollar into a spiralling hyperinflation. The dollars would sooner or later end up in the United States, the only country where people are forced to accept them because of their legal-tender status. Here all

prices would soar, possibly entailing a hyperinflation and collapse of the entire monetary system.

The same considerations apply, *mutatis mutandis*, to all other international paper-money standards. The point is that, in the present regime of virtually unhampered international flows of capital, it is out of the question to prevent the outflow of standard money into foreign countries. And the more of that money that accumulates abroad, the more its producer risks being subject to the sort of blackmail we have already discussed.⁷ Notice the irony that the potential for such blackmail is greatest precisely when the institutional safeguards against fluctuating exchange rates are strongest—in the case of currency boards and dollarization.

The leadership of the U.S. Federal Reserve is aware of this situation. To guard itself against the danger of switching, it has developed a program of shared seignorage. That is, U.S. authorities actually pay foreign governments for dollarizing their economies, and especially for maintaining the dollarization.

However, such schemes for the integration of standard monies and secondary currencies have been applied, so far, only in relatively unimportant cases. The only realistic scenario that could curb the expansionary drift of monetary blackmail as analyzed above is cooperation between the producers of standard paper money. For example, if in a dollar crisis the euro producers commit to stabilize the dollar-euro exchange rate on the downside, then the financial incentives for going out of dollar-denominated assets and into euro-denominated assets would largely disappear.

But why should producers of standard money such as the euro be willing to assist a competitor in dire straits? There are

⁷The same thing would happen on a national scale if there were competing central banks. In times of strain on the reserves, the commercial banks could then threaten to switch from one central bank to another, dooming in the process the system they leave. This is one of the reasons why no bank has ever assumed the responsibilities of a central bank (lender of last resort) without being compensated through a legal monopoly that prevented such switching. The Bank of England is a case in point.

at least two good reasons for such cooperation. First, they might wish to discourage monetary blackmail by the producers of secondary currencies, because in the next round they themselves could be the victims of such attempts. Second, they themselves would be negatively affected in the event of a currency crisis hitting their competitor. It is true that in the short run they would benefit from investors rushing into euro-denominated assets. However, they could not prevent that the same investors rush out again once the dollar-crisis has been solved, for example, through some monetary reform. Standard paper-money producers would thus be ill-advised to play cat and mouse with international investors, in the hope to profit from a currency crisis hitting one of their competitors.

Now the crucial point is that all relevant parties know all this and that therefore moral hazard comes into play again. Paper-money producers have a strong incentive to expand their production because they know that their competitors, acting in their own interest, would be likely to assist them whenever they are threatened with a currency crisis. Thus we find the same strong incentive for expansionary collusion between paper-money producers that we have already described in earlier sections for the case of domestic fractional-reserve banks. This monetary expansion path results from the very nature of paper-money competition, just as the expansion of fractional-reserve certificates results from the very nature of competitive fractional-reserve banking.

Is there any way out of this monetary quagmire? One solution would be the return to autarky, cutting all ties with the international currency and financial markets; but this would entail misery and starvation, and is therefore not really an option. Another solution would be to merge the standard paper money producers, possibly along the lines of the European System of Central Banks and possibly along with international regulation of capital markets and the banking industry.⁸ But is world paper-money union a viable solution?

⁸See Stephen F. Frowen, "The Functions of Money and Financial Credit: Their Objectives, Structure and Inbuilt Deficiencies," *Journal of the Association of Christian Economists* 14 (February 1993).

4. DEAD END OF THE WORLD PAPER-MONEY UNION

As we have seen, there is a strong tendency for the formation of currency blocks around the paper monies used in the countries with the largest capital markets. The driving force in this process is the quest of foreign governments for additional revenue. The governments that control the large capital markets have little incentive to adopt the currencies controlled by other governments. But governments that control only a small tax base and cannot tame their appetite for more money must at some point turn to international capital markets; and this sooner or later forces them to adopt a foreign paper money, or to merge its paper money with the paper monies controlled by other governments.

We have also seen that the connectivity between international capital markets creates an incentive for competing standard paper-money producers to cooperate and, eventually, to merge. This consolidation and centralization process is at present far from being completed. Today's international monetary order is an order in transition. In the preceding section we have analyzed some of the problems that could manifest themselves in the next few years if political leaders do not take appropriate action. We have pointed out that one way of avoiding a world of spiraling hyperinflations and currency wars is global monetary integration on a paper standard. There would then be just one paper money for the entire world, possibly with a few remaining national paper currencies that serve as money certificates for the global money. The great project that Lord Keynes unsuccessfully promoted at the 1944 Bretton Woods conference would then finally have come true.

We have already pointed out all essential implications of such an event. Even a national paper money is a powerful engine of economic, cultural, and spiritual degradation. How much more would this be the case with a global paper money? Such a monetary regime would provide the economic foundation of a totalitarian nightmare.

It is true that we are still far away from this scenario. Great obstacles stand in its way, because it would require no less

than a political unification of mankind. But let us assume for the moment that these problems could be overcome in the near future. And let us also assume that fears of totalitarianism could be dispelled by an appropriate moral education of political leaders, who would then excel in the art of self-restraint. Would this solve the problem of monetary constitution? Would it give the world a true monetary order that did not bear in its very bosom a tendency for self-destruction, a tendency inherent in all fiat monetary systems?

In light of our general discussion of paper money, the answer is patent. All paper-money systems, be they national or international, labor under the presence of moral hazard. In the long run, therefore, a global paper money cannot evade the fate of national paper money. It must either collapse in hyperinflation or force the government to adopt a policy of increasing control, and eventually total control, over all economic resources. Both scenarios entail economic disruptions on a scale that we can barely imagine today. The inevitable result would be death for many hundreds of millions of human beings.

There is hope, however. Mankind is free to return at any time to the natural production of money, which is in fact the only ethically justifiable and economically viable monetary order.

Topic 1: Globalization and the Multinational Firm

We take a look inside the 2021 Annual Report published by Belden Inc. (ticker symbol BDC) to determine the extent of the international financial risk exposure and management.

We will conduct word searches in a PDF file of the Annual Report document for the following terms:

- “international”
- “currency” (to pick up references to foreign currency)
- “hedg” (to pick up references to both “hedge” and “hedging”)
- “derivative”

In the pages that follow are some examples from our word search through the 2021 Annual Report of Belden Inc.



Customers

We sell to distributors, OEMs, installers, and end-users. For the year ended December 31, 2021, sales to our largest distributor represented approximately 16% of our consolidated revenues. No other customer accounted for more than 10% of our revenues in 2021.

We have supply agreements with distributors and OEM customers. In general, our customers are not contractually obligated to buy our products exclusively, in minimum amounts, or for a significant period of time. We believe that our relationships with our customers and distributors are good and that they are loyal to Belden products as a result of our reputation, the breadth of our product portfolio, the quality and performance characteristics of our products, and our customer service and technical support, among other reasons.

International Operations

In addition to manufacturing facilities in the United States (U.S.), we have manufacturing and other operating facilities in Canada, China, India, Mexico, and St. Kitts, as well as various countries in Europe. During 2021, approximately 47% of Belden's sales were to customers outside the U.S. Our primary channels to international markets include both distributors and direct sales to end users and OEMs. Financial information for Belden by country is shown in Note 6 to the Consolidated Financial Statements.

Competition

The markets in which we operate can be generally categorized as highly competitive with many players. In order to maximize our competitive advantages, we manage our product portfolio to capitalize on secular trends and high-growth applications in those markets. Based on available data for our served markets, we estimate that our market share across our segments is significant, ranging from approximately 5% – 20%. A substantial acquisition in one of our served markets would be necessary to meaningfully change our estimated market share percentage.

The principal competitive factors in all our product markets are technical features, quality, availability, price, customer support, and distribution coverage. The relative importance of each of these factors varies depending on the customer. Some products are manufactured to meet published industry specifications and are less differentiated on the basis of product characteristics. We believe that Belden stands out in many of our markets on the basis of the breadth of our product portfolio, the quality and performance characteristics of our products, our customer service, and our technical support.

Research and Development

We conduct research and development on an ongoing basis, including new and existing hardware and software product development, testing and analysis, and process and equipment development and testing. See the Consolidated Statements of Operations for amounts incurred for research and development. Many of the markets we serve are characterized by advances in information processing and communications capabilities, including advances driven by the expansion of digital technology, which require increased transmission speeds and greater bandwidth. Our markets are also subject to increasing requirements for mobility, information security, and transmission reliability. We believe that our future success will depend in part upon our ability to enhance existing products and to develop, manufacture and deliver new products that meet or anticipate such changes in our served markets.

In our Industrial Solutions segment, customers are rapidly adopting new technology to enable digital transformations and improve their Environmental, Social & Governance (ESG) impact. This includes deploying Industry 4.0 to increase visibility of their digitized assets and adopting Artificial Intelligence (AI) to increase analytics and autonomous decision-making in their systems. These approaches need users to refine workflows by collecting data from disparate sources, transmitting it to points of consolidation and decision making, and converting it to standard formats that application software can use. This overall process can be referred to as "digitization" and a key part of our research and development is focused on supporting these customer journeys with technology that add value at multiple steps in the digitization process, during data acquisition, data transmission, and data orchestration and management. Our research and development enables customized enhanced solutions to support customers' innovative methods surrounding the collection, analysis, and transmission of data.

Additionally, in our Industrial Solutions segment, there is a compelling need among customers to detect, prevent and respond to cyber security threats. This is a long-standing need within corporate networks, but we believe the rapid proliferation of new

The effects of the COVID-19 pandemic continued to materially affect how we and our customers operated our businesses in 2021, and the duration and extent to which this or future epidemics, pandemics or other major disasters will impact our future results of operations and overall financial performance remains uncertain.

In December 2019, a novel coronavirus disease (“COVID-19”) was first reported and on March 11, 2020, the World Health Organization characterized COVID-19 as a pandemic that has yet to fully recede. The widespread health crisis is adversely affecting the broader economies, financial markets and may adversely affect the overall demand environment for many of our products.

Our operations and the operations of our suppliers, channel partners and customers were and continue to be disrupted to varying degrees by a range of external factors related to the COVID-19 pandemic, some of which are not within our control. Many governments imposed, and may yet impose or may re-impose, a wide range of restrictions on the physical movement or congregation of people in order to limit the spread of COVID-19. The COVID-19 pandemic has had, and likely will continue to have, an impact on the attendance and productivity of our employees, and those of our channel partners or customers, resulting in negative impacts to our results of operations and overall financial performance. Additionally, COVID-19 has resulted, and may result in future periods, in delays in non-residential construction, non-crisis-related IT purchases and project completion schedules in general, all of which can negatively impact our results in both current and future periods.

The duration and extent of the impact from the COVID-19 pandemic or any future epidemic, pandemic or major disaster depends on future developments that cannot be accurately predicted at this time, such as the severity and transmission rate of the virus (including variant mutations of the virus), the extent and effectiveness of containment actions, treatments and vaccinations, the effects of measures enacted by policy makers and central banks around the globe, and the impact of these and other factors on our employees, customers, channel partners and suppliers. If we are not able to respond to and manage the impact of such events effectively, our business will be affected.

Our results of operations are subject to foreign and domestic political, social, economic, and other uncertainties and are affected by changes in currency exchange rates.

In addition to manufacturing and other operating facilities in the U.S., we have manufacturing and other operating facilities in Canada, China, India, Mexico, St. Kitts, and several European countries. We rely on suppliers in many countries, including China. Our foreign operations are subject to economic, social, and political risks inherent in maintaining operations abroad such as economic and political destabilization, land use risks, international conflicts, pandemics and other health-related crises, restrictive actions by foreign governments, and adverse foreign tax laws. In addition to economic and political risk, a risk associated with our European manufacturing operations is the higher relative expense and length of time required to adjust manufacturing employment capacity. We also face political risks in the U.S., including tax or regulatory risks or potential adverse impacts from legislative impasses over, or significant legislative, regulatory or executive changes in fiscal or monetary policy and other foreign and domestic government policies, including, but not limited to, trade policies and import/export policies.

Approximately 47% of our sales are outside the U.S. Other than the U.S. dollar, the principal currencies to which we are exposed through our manufacturing operations, sales, and related cash holdings are the euro, the Canadian dollar, the Hong Kong dollar, the Chinese yuan, the Mexican peso, the Australian dollar, the British pound and Indian rupee. Generally, we have revenues and costs in the same currency, thereby reducing our overall currency risk, although any realignment of our manufacturing capacity among our global facilities could alter this balance. When the U.S. dollar strengthens against other currencies, the results of our non-U.S. operations are translated at a lower exchange rate and thus into lower reported revenues and earnings.

A challenging global economic environment or a downturn in the markets we serve could adversely affect our operating results and stock price in a material manner.

A challenging global economic environment could cause substantial reductions in our revenue and results of operations as a result of weaker demand by the end users of our products and price erosion. Price erosion may occur through competitors becoming more aggressive in pricing practices. A challenging global economy could also make it difficult for our customers, our vendors, and us to accurately forecast and plan future business activities. Our customers could also face issues gaining timely access to sufficient credit, which could have an adverse effect on our results if such events cause reductions in revenues, delays in collection, or write-offs of receivables. Further, the demand for many of our products is economically sensitive and will vary with general economic activity, trends in nonresidential construction, investment in manufacturing facilities and automation, demand for information technology equipment, and other economic factors.

the lesser of 10% of the projected benefit obligation or the fair market value of plan assets, amortized over the estimated future working life of the plan participants.

As a sensitivity measure, the effect of a 50 basis point decline in the assumed discount rate would have resulted in an increase in the 2021 net periodic benefit cost and projected benefit obligations as of December 31, 2021 of approximately \$2.0 million and \$38.0 million, respectively. A 50 basis point decline in the expected return on plan assets would have resulted in an increase in the 2021 net periodic benefit cost of approximately \$1.8 million.

Conversely, the effect of a 50 basis point increase in the assumed discount rate would have resulted in a decrease in the 2021 net periodic benefit cost of approximately \$1.6 million and a decrease in the projected benefit obligation of approximately \$34.2 million as of December 31, 2021. A 50 basis point increase in the expected return on plan assets would have resulted in a decrease in the 2021 net periodic benefit cost of approximately \$1.8 million.

Acquisition Accounting

We allocate the consideration of an acquired business to its identifiable assets and liabilities based on estimated fair values. The excess of the consideration over the amount allocated to the assets and liabilities, if any, is recorded to goodwill. We use all available information to estimate fair values. We typically engage third party valuation specialists to assist in the fair value determination of inventories, tangible long-lived assets, and intangible assets other than goodwill. The carrying values of acquired receivables and accounts payable have historically approximated their fair values as of the acquisition date. As necessary, we may engage third party specialists to assist in the estimation of fair value for certain liabilities. We adjust the preliminary acquisition accounting, as necessary, typically up to one year after the acquisition closing date as we obtain more information regarding asset valuations and liabilities assumed.

Our acquisition accounting methodology contains uncertainties because it requires management to make assumptions and to apply judgment to estimate the fair value of acquired assets and liabilities. Management estimates the fair value of assets and liabilities based upon quoted market prices, the carrying value of the acquired assets and widely accepted valuation techniques, including discounted cash flows and market multiple analyses. Unanticipated events or circumstances may occur which could affect the accuracy of our fair value estimates, including assumptions regarding industry economic factors and business strategies.

If actual results are materially different than the assumptions we used to determine fair value of the assets and liabilities acquired through a business combination, it is possible that adjustments to the carrying values of such assets and liabilities will have an impact on our net earnings. See Note 4.

Item 7A. Quantitative and Qualitative Disclosures About Market Risk

Market risks relating to our operations result primarily from currency exchange rates, certain commodity prices, interest rates, and credit extended to customers. Each of these risks is discussed below.

Currency Exchange Rate Risk

We are exposed to foreign currency risks that arise from normal business operations. These risks include the translation of local currency balances of foreign subsidiaries and transactions denominated in currencies other than a location's functional currency.

Our investments in certain foreign subsidiaries are recorded in currencies other than the U.S. dollar. As these foreign currency denominated investments are translated at the end of each period during consolidation using period-end exchange rates, fluctuations of exchange rates between the foreign currency and the U.S. dollar increase or decrease the value of those investments. These fluctuations and the results of operations for foreign subsidiaries, where the functional currency is not the U.S. dollar, are translated into U.S. dollars using the average exchange rates during the year, while the assets and liabilities are translated using period end exchange rates. The assets and liabilities-related translation adjustments are recorded as a separate component of accumulated other comprehensive income (loss) in our Consolidated Balance Sheets. We generally view our investments in international subsidiaries with functional currencies other than the U.S. dollar as long-term. As a result, we do not generally use derivatives to manage these net investments. However, we designated euro debt issued by Belden Inc., a USD functional currency entity, as a net investment ~~hedge~~ of certain international subsidiaries. See Note 16 for further discussion.

Transactions denominated in currencies other than a location's functional currency may produce receivables or payables that are fixed in terms of the amount of foreign currency that will be received or paid. A change in exchange rates between the functional currency and the currency in which a transaction is denominated increases or decreases the expected amount of functional currency cash flows upon settlement of the transaction. That increase or decrease in expected functional currency cash flows is a foreign exchange transaction gain or loss that is included in our operating income in the Consolidated Statements of Operations. In 2021, we recorded approximately \$1.7 million of net foreign currency transaction losses. In 2020, we recorded approximately \$7.9 million of net foreign currency transaction gains.

Generally, the currency in which we sell our products is the same as the currency in which we incur the costs to manufacture our products, resulting in a natural hedge. Our currency exchange rate management strategy primarily involves the use of natural techniques, where possible, such as the offsetting or netting of like-currency cash flows. However, we re-evaluate our strategy as the foreign currency environment changes, and it is possible that we could utilize derivative financial instruments to manage this risk in the future. We did not have any foreign currency derivatives outstanding as of December 31, 2021.

Our exposure to currency rate fluctuations primarily relates to exchange rate movements between the U.S. dollar and the euro, Canadian dollar, Hong Kong dollar, Chinese yuan, Mexican peso, Australian dollar, British pound, and Indian rupee.

Commodity Price Risk

Certain raw materials used by us are subject to price volatility caused by supply conditions, political and economic variables, and other unpredictable factors. The primary purpose of our commodity price management activities is to manage the volatility associated with purchases of commodities in the normal course of business. We do not speculate on commodity prices.

We are exposed to price risk related to our purchase of copper used in our products, although we are generally able to raise selling prices to customers to cover the increase in copper costs. Our copper price management strategy involves the use of natural techniques, where possible, such as purchasing copper for future delivery at fixed prices. We do not generally use commodity price derivatives and did not have any outstanding at December 31, 2021 or 2020. The following table presents unconditional commodity purchase obligations outstanding as of December 31, 2021. The unconditional purchase obligations will settle during 2022.

	<u>Purchase Amount</u>	<u>Fair Value</u>
	<u>(In thousands, except average price)</u>	
Unconditional copper purchase obligations:		
Commitment volume in pounds	4,766	
Weighted average price per pound	<u>\$ 4.38</u>	
Commitment amounts	<u>\$ 20,889</u>	<u>\$ 21,234</u>

We are also exposed to price risk related to our purchase of selected commodities derived from petrochemical feedstocks used in our products. We generally purchase these commodities based upon market prices established with the vendors as part of the purchase process. Pricing of these commodities is volatile as they tend to fluctuate with the price of oil. Historically, we have not used commodity financial instruments to hedge prices for commodities derived from petrochemical feedstocks.

Belden Inc.
Consolidated Cash Flow Statements

Years Ended December 31,

	2021	2020	2019
	(In thousands)		
Cash flows from operating activities:			
Net income (loss)	\$ 64,317	\$ (55,058)	\$ (376,776)
Adjustments to reconcile net income (loss) to net cash provided by operating activities:			
Depreciation and amortization	87,988	108,687	139,259
Goodwill and other asset impairment	140,461	113,007	521,441
Share-based compensation	24,871	20,030	17,751
Loss on debt extinguishment	5,715	—	—
Deferred income tax expense (benefit)	3,575	(19,410)	(23,540)
Changes in operating assets and liabilities, net of the effects of exchange rate changes, acquired businesses, and disposals:			
Receivables	(119,012)	70,707	22,926
Inventories	(92,984)	(8,507)	44,477
Accounts payable	135,666	(43,567)	(41,527)
Accrued liabilities	61,241	7,374	(17,654)
Income taxes	(6,448)	(22,823)	5,497
Other assets	(12,693)	2,018	(16,118)
Other liabilities	(20,642)	906	1,157
Net cash provided by operating activities	272,055	173,364	276,893
Cash flows from investing activities:			
Capital expenditures	(90,982)	(90,215)	(110,002)
Cash from (used for) business acquisitions, net of cash acquired	(73,340)	590	(74,392)
Purchase of intangible assets	(3,650)	—	—
Proceeds from disposal of tangible assets	30,234	3,161	25
Proceeds from disposal of businesses, net of cash sold	45,735	54,821	—
Net cash used for investing activities	(92,003)	(31,643)	(184,369)
Cash flows from financing activities:			
Payments under borrowing arrangements	(360,304)	(190,000)	—
Cash dividends paid	(9,056)	(9,029)	(34,439)
Debt issuance costs paid	(8,173)	—	—
Withholding tax payments for share based-payment awards	(5,570)	(1,388)	(2,149)
Payments under financing lease obligations	(3,151)	(194)	(360)
Payments to noncontrolling interest holders	(2,682)	—	—
Payments under share repurchase program	—	(35,000)	(50,000)
Payment of earnout consideration	—	(29,300)	—
Borrowings under credit arrangements	356,010	190,000	—
Net cash used for financing activities	(32,926)	(74,911)	(86,948)
Effect of foreign currency exchange rate changes on cash and cash equivalents	(5,363)	9,299	(301)
Increase in cash and cash equivalents	141,763	76,109	5,275
Cash and cash equivalents, beginning of year	501,994	425,885	420,610
Cash and cash equivalents, end of year	\$ 643,757	\$ 501,994	\$ 425,885

The Consolidated Cash Flow Statement includes the results of discontinued operations up to the disposal date, July 2, 2020. The accompanying notes are an integral part of these Consolidated Financial Statements.

Notes to Consolidated Financial Statements

Note 1: Basis of Presentation

Business Description

Belden Inc. (the Company, us, we, or our) is a global supplier of specialty networking solutions built around two global businesses – Enterprise Solutions and Industrial Solutions. Our comprehensive portfolio of solutions enables customers to transmit and secure data, sound, and video for mission critical applications across complex enterprise and industrial environments.

Consolidation

The accompanying Consolidated Financial Statements include Belden Inc. and all of its subsidiaries, including variable interest entities for which we are the primary beneficiary. We eliminate all significant affiliate accounts and transactions in consolidation.

Foreign Currency

For international operations with functional currencies other than the United States (U.S.) dollar, we translate assets and liabilities at current exchange rates; we translate income and expenses using average exchange rates. We report the resulting translation adjustments, as well as gains and losses from certain affiliate transactions, in accumulated other comprehensive income (loss), a separate component of stockholders' equity. We include exchange gains and losses on transactions in operating income.

We determine the functional currency of our foreign subsidiaries based upon the currency of the primary economic environment in which each subsidiary operates. Typically, that is determined by the currency in which the subsidiary primarily generates and expends cash. We have concluded that the local currency is the functional currency for all of our material subsidiaries.

Reporting Periods

Our fiscal year and fiscal fourth quarter both end on December 31. Our fiscal first quarter ends on the Sunday falling closest to 91 days after December 31. Our fiscal second and third quarters each have 91 days.

Use of Estimates in the Preparation of the Financial Statements

The preparation of financial statements in conformity with accounting principles generally accepted in the U.S. requires us to make estimates and assumptions that affect the reported amounts of assets, liabilities, and operating results and the disclosure of contingencies. Actual results could differ from those estimates. We make significant estimates with respect to the collectability and valuation of receivables, the valuation of inventory, the realization of deferred tax assets, the valuation of goodwill and indefinite-lived intangible assets, the valuation of contingent liabilities, the calculation of share-based compensation, the calculation of pension and other postretirement benefits expense, and the valuation of acquired businesses.

Note 2: Summary of Significant Accounting Policies

Fair Value Measurement

Accounting guidance for fair value measurements specifies a hierarchy of valuation techniques based upon whether the inputs to those valuation techniques reflect assumptions other market participants would use based upon market data obtained from independent sources or reflect our own assumptions of market participant valuation. The hierarchy is broken down into three levels based on the reliability of the inputs as follows:

- Level 1 – Quoted prices in active markets that are unadjusted and accessible at the measurement date for identical, unrestricted assets or liabilities;
- Level 2 – Quoted prices for identical assets and liabilities in markets that are not active, quoted prices for similar assets and liabilities in active markets, or financial instruments for which significant inputs are observable, either directly or indirectly; and
- Level 3 – Prices or valuations that require inputs that are both significant to the fair value measurement and unobservable.

Topic 2: How to Read Foreign Exchange Rates

Reading foreign exchange rates can be confusing when you don't know what the terms and numbers mean. What is a currency pair? Which currency is being quoted in terms of the other currency? What is the difference between the bid and ask price? These are some of the questions you can only answer once you have learned to read foreign exchange rates correctly.

Every currency has a name ("dollar"), a three-letter currency code (USD), and a symbol (\$) that can be placed in front of a numerical value (\$1.50). Foreign exchange quotes always use the three-letter currency codes. A list of foreign currencies by country (and currency name): <https://www.xe.com/symbols.php>

Currency Pair

Currencies are quoted in pairs of one currency valued by another. In other words, a currency quote compares the value of one currency against another one. One of these currencies is listed in the quote as the "base" currency, and the other is the "quoted" currency. The quote itself (the number) shows how much of the "quoted" currency is needed to buy one unit of the "base" currency.

The currency exchange rates listed in the table on the opposite page are taken from: <https://www.investing.com/currencies/streaming-forex-rates-majors>.

The first quote listed is the exchange rate between the Euro (EUR) and the US Dollar (USD). The pair is expressed as **EUR/USD**. The currency code on the left (EUR) is the "base" currency and the code on the right (USD) is the "quoted" currency. The base currency is always fixed at one unit, while the quoted currency is the price of one unit the other currency.

Bid and Ask

When trading currencies, it's important to know which price is appropriate for the transaction. There are two price quotes: **Bid price** and the **Ask price**. These terms are used from the standpoint of the bank (the dealer) that is executing the foreign currency exchange transaction. Simply put, the Bid is the Buy price (i.e., the price the bank will pay to buy it from you), while the Ask is the Sell price (the price at which the bank will sell it to you). The difference between the Buy and Ask prices is called the "spread" and represents a profit for foreign currency dealers. If you want to purchase foreign currency with USD from a dealer, you must look at the Ask price. On the other hand, if you want to sell foreign currency to a dealer and receive USD, you must look at the Bid price. Thus, based on that first quote on the opposite page, the bank will buy Euros from a seller at \$1.1028 and will sell Euros to a buyer at \$1.1029 (note that the spread is very small: \$0.0001).

Direct Quote vs Indirect Quote

There are two types of currency quotes – direct and indirect. A direct quote is where the domestic currency is the quoted currency (i.e., the foreign currency remains fixed at one unit and its cost in USD changes). With indirect quote, the domestic currency is fixed at one unit while the foreign currency changes. Notice which currency pairs with USD are Direct quotes (e.g., EUR and GBP) and which are Indirect quotes (e.g., JPY, TRY, CHF, CAN). Notice also the quotes without USD.

Cross Currency

Some currency quotes do not have a U.S. dollar in the pair. This is called a cross-currency rate for Americans, and this allows one foreign currency to be traded for another without the need to exchange the currency into U.S. dollars. Cross currency pairs give you more trading options, but the caveat here is that these pairs are not as popular as the ones that involve the U.S. dollar. We learn more about cross-currency rates later in the semester.

Country and Currency	Currency Code	Graphic Image
Albania Lek	ALL	Lek
Afghanistan Afghani	AFN	ؔ
Argentina Peso	ARS	\$
Aruba Guilder	AWG	f
Australia Dollar	AUD	\$
Azerbaijan Manat	AZN	₼
Bahamas Dollar	BSD	\$
Barbados Dollar	BBD	\$
Belarus Ruble	BYN	Br
Belize Dollar	BZD	BZ\$
Bermuda Dollar	BMD	\$
Bolivia Boliviano	BOB	\$b
Bosnia and Herzegovina Convertible Mark	BAM	KM
Botswana Pula	BWP	P
Bulgaria Lev	BGN	лв
Brazil Real	BRL	R\$

Simple Rules for Currency Exchange Rates

1. **Direct quote** = when rate contains your home country's currency symbol

Example: Americans quoting the USD exchange rate with the British Pound

GBPUSD or **GBP/USD** quote = **\$1.2946** = \$ to £1

2. **Indirect quote** = when rate contains the other country's currency symbol

Example: Americans quoting the USD exchange rate with the British Pound

USDGBP or **USD/GBP** quote = **£0.7724** = £ to \$1

Note: The **direct** and **indirect** quotes are reciprocals (i.e., multiplicative **inverses**) of one another

Note: The reciprocal or inverse of "x" = "1 ÷ x" thus, in the example above, the reciprocal of \$1.2946 = is (1 ÷ \$1.2946) = £0.7724

3. When converting a FOREX position, you need two elements: the **amount** and the **exchange rate**

Rule: When both symbols are the same = divide

Example: Converting an amount of USD into British pounds

<u>Amount</u>		<u>Exch. Rate</u>		<u>Result</u>
\$500,000	÷	\$1.2946	=	£386,220

Example: Converting an amount of British pounds into USD

£500,000	÷	£0.7724	=	\$647,300
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Rule: When both symbols are different = multiply

Example: Converting an amount of USD into British pounds

\$500,000	x	£0.7724	=	£386,220
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Example: Converting an amount of British pounds into USD

£500,000	x	\$1.2946	=	\$647,300
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4. When describing currency movements, you must identify three facts:

- a. Which currency moved?
- b. Which currency did it move against?
- c. What was the result (depreciation or appreciation) of the currency?

Example (direct quote):

Exchange rate of GBPUSD on **1-1-2024** **\$1.2946**

Exchange rate of GBPUSD on **7-1-2024** **\$1.3574**

- a. Which currency moved? **The British pound (GBP)**
- b. Which currency did it move against? **The US dollar (USD)**
- c. What was the result (depreciation or appreciation)?

Fact: One GBP purchased more USD on 7/1 than on 1/1

Fact: It required more USD to purchase one GBP on 7/1 than on 1/1

Answer: **The GBP appreciated against the USD by 4.85%**

Note: % change = (End Rate - Beg. Rate) ÷ Beg. Rate

Thus, % change = (1.3574 - 1.2946) ÷ 1.2946 = .0485 = 4.85%

Note: The above problem could also be correctly stated as the USD moving against the GBP

Answer: **The USD depreciated against the GBP by 4.85%**

Topic 3: Foreign Exchange Rates

Spot and Forward

The most common type of foreign exchange transaction is for immediate exchange. The market where these transactions occur is known as the **spot market**, and the exchange rate at which one currency is traded for another in the spot market is known as the **spot rate**. Unless otherwise noted, when one exchange rate is quoted, it is usually the **ask price**.

For large transactions, the spot rate is usually quoted for settlement on the **transaction date + 2 business days**.

Forward exchange rates are quoted at **premiums (+)** or **discounts (-)** to the spot rate. These premiums and discounts are useful insights for analysts to gauge what to expect from the market, and which currencies are expected to appreciate and which ones are expected to depreciate.

The forward exchange rates are quoted in terms of points (also known as “pips” which stands for “points in percentage”).

For example, let's say the current EUR/USD exchange rate is 1.2823. The forward quote (ask price) for a 90-day forward exchange rate is +16 points (quoted as 16.0000). This 16 points will be interpreted as $16 \times (1/10,000) = 0.0016$ above the spot rate. A positive sign means that euro is trading at a premium relative to US dollar.

The outright forward quote would be converted to this:

$$1.2823 \text{ (spot)} + 0.0016 \text{ (forward pip)} = 1.2839 \text{ (i.e., 1 EUR = \$1.2839)}$$

If the forward quote was -15 points instead, this means that the euro is trading at a discount relative to US dollar.

$$\text{The outright forward quote would be} = 1.2823 - 0.0015 = 1.2808$$

We can also express the forward premiums and discounts in terms of percentages of the spot rate at the time of the forward-rate quote. For example, the premium of 16 points explained above can be expressed as $0.0016/1.2823 = 0.0012$ or 0.12%.

Let's take one more example. The current USD/SGD spot rate is 1.2848. The quoted 1 year forward rate is -5.5 points.*

The outright forward exchange rate will be $1.2848 - 0.00055 = 1.28425$, which means it is a discount of .00429 or -0.429%.

***Note that the forward points have a negative sign, which means that the Singapore dollar (SGD) is trading at a forward discount relative to USD.**

Symbol	Description
TOM	Tomorrow; this Value Date is not available for the USD/CAD currency pair.
Spot	Transaction date (today) + 2 business days. If CAD is the base currency, transaction date (today) +1 business day.
T3	Transaction date (today) + 3 business days.
TN	1 day.
ON	Overnight (1 day).
SN	Spot Next
1D	1 day from the transaction date (today).
2D	2 days from the transaction date (today).
3D	3 days from the transaction date (today).
1W	1 week from the transaction date (today).
1M	1 month from the transaction date (today).
2M	2 months from the transaction date (today).
3M	3 months from the transaction date (today).
6M	6 months from the transaction date (today).
9M	9 months from the transaction date (today).
1Y	1 year from the transaction date (today).
IMM1	The nearest International Monetary Market delivery date.
IMM2	The subsequent International Monetary Market delivery date.

The schedule of Forward Rates for the EUR/USD on the following two pages is taken from the following website: <https://www.investing.com/>

From the horizontal menu, pull down the Markets submenu and click Currencies, then click on the EUR/USD page, and then the option for Forward Rates.

Topic 4: Foreign Currency Transaction Exposure and Forward Rates

In-Class Exercise

In-Class Case: PAYABLE

Ravencorp is a distributor of sports apparel. Its products are marketed all over the U.S. and many parts of Europe as well. When the company buys merchandise from foreign suppliers, it generally pays in USD in order to guard against adverse exchange rate movements.* However, today it placed an order with a large supplier in central England for £1,000,000 with the condition that delivery be made in two months (180 days) and payment made in Pounds (GBP).*

You book an Account Payable (A/P) in USD. Your risk is that the GBP will appreciate versus the USD over the next two months, thus increasing the cost when the GBP payable is settled.* You decides to contact the company's bank and ask for suggestions about hedging (i.e., insuring against) the exchange rate exposure.* The bank sends you a schedule of Spot and Forward Rates for the GBPUSD (**see spot and forward schedule below**) and offers to set up a forward contract for purchasing the GBP needed to make the payable in 180 days (6 months). **Note: You must first determine whether the appropriate exchange rate for this Payable is the Bid or the Ask.**

Complete in Excel a profile of this transaction that includes the following:

1. The Dates of Transaction and Settlement.
2. The Spot Rate* (direct quote) and the Book Value of the Payable (in USD) on the Date of Transaction
3. A table of data and a graph of the Risk Profile of this Payable
4. The six-month Forward Rate* (direct quote)
5. The forward premium/discount (expressed as a %) of the GBP based on the forward rate
6. The gain/loss on the A/P (expressed monetarily) based on the forward rate
7. The gain/loss on the A/P (expressed monetarily) if you had remained unhedged and the GBP had in reality depreciated by 1.00% from transaction (booking) date to settlement date.

*Discuss the issue for each asterisk, consider the controller's alternatives, make a recommendation.

In-Class Case: RECEIVABLE

Also today, Ravencorp received an order from a large retailer in Germany for €1,000,000 of its products, with delivery in two months (60 days) and payment in EUR. You book an Accounts Receivable (A/R). Now your risk is that the EUR might depreciate against the USD over the next two months, thus eliminating some or most of the profit when the A/R (in EUR) is collected and converted to USD in 180 days (6 months). You contact the bank (**see spot and forward schedule below**). **Note: You must again first determine whether the appropriate exchange rate for this Payable is the Bid or the Ask.**

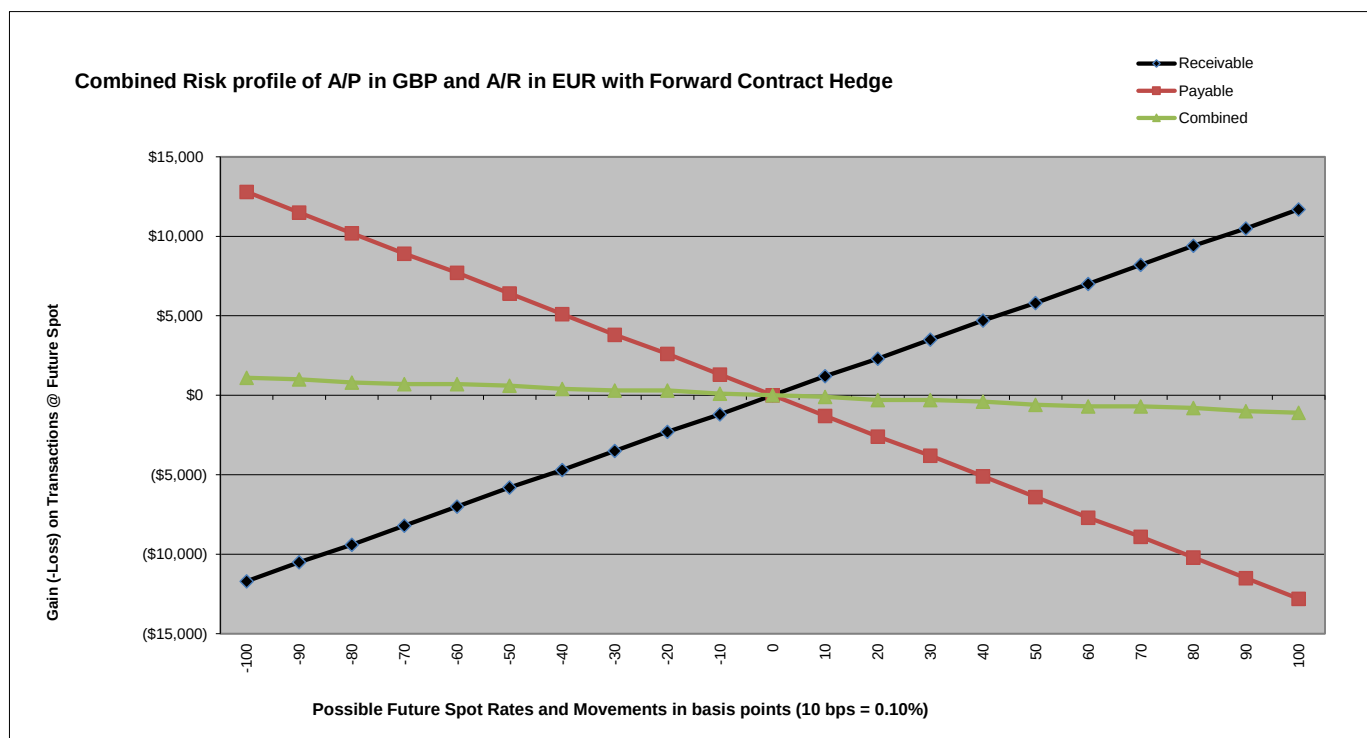
Repeat the process from the Payable example above:

1. The Dates of Transaction and Settlement.
2. The Spot Rate* (direct quote) and the Book Value of the Payable (in USD) on the Date of Transaction
3. A table of data and a graph of the Risk Profile of this Payable
4. The six-month Forward Rate* (direct quote)
5. The forward premium/discount (expressed as a %) of the EUR based on the forward rate
6. The gain/loss on the A/R (expressed monetarily) based on the forward rate
7. The gain/loss on the A/R (expressed monetarily) if you had remained unhedged and the EUR had in reality appreciated by 1.00% from transaction (booking) date to settlement date.

Note: A graph of the "combined profile is on the opposite page. The pages that follow will provide information on the two individual profiles (i.e., the A/P and the A/R).

Combined A/P (GBP) and A/R (EUR) : Original Position (Short and Long)

Movement Basis Points	Movement as a Percent	Gain (Loss) on A/P	Gain (Loss) on A/R	Net Gain (Loss) on both
-100	-1.00%	\$12,800	(\$11,700)	\$1,100
-90	-0.90%	\$11,500	(\$10,500)	\$1,000
-80	-0.80%	\$10,200	(\$9,400)	\$800
-70	-0.70%	\$8,900	(\$8,200)	\$700
-60	-0.60%	\$7,700	(\$7,000)	\$700
-50	-0.50%	\$6,400	(\$5,800)	\$600
-40	-0.40%	\$5,100	(\$4,700)	\$400
-30	-0.30%	\$3,800	(\$3,500)	\$300
-20	-0.20%	\$2,600	(\$2,300)	\$300
-10	-0.10%	\$1,300	(\$1,200)	\$100
0	0.00%	\$0	\$0	\$0
10	0.10%	(\$1,300)	\$1,200	(\$100)
20	0.20%	(\$2,600)	\$2,300	(\$300)
30	0.30%	(\$3,800)	\$3,500	(\$300)
40	0.40%	(\$5,100)	\$4,700	(\$400)
50	0.50%	(\$6,400)	\$5,800	(\$600)
60	0.60%	(\$7,700)	\$7,000	(\$700)
70	0.70%	(\$8,900)	\$8,200	(\$700)
80	0.80%	(\$10,200)	\$9,400	(\$800)
90	0.90%	(\$11,500)	\$10,500	(\$1,000)
100	1.00%	(\$12,800)	\$11,700	(\$1,100)



GBP/USD - British Pound US Dollar

 Real-time FX

1.2756

+0.0025 (+0.20%)

09:05:56 - Real-time Data. ([Disclaimer](#))

Add to Portfolio

Create Alert

Type:

Group:

Base:

Second:

Currency

Major

British Pound

US Dollar

Prev. Close: 1.2731 Bid/Ask: 1.2754 / 1.2758 Day's Range: 1.2676 - 1.2757

- General Chart News & Analysis Technical Forum Brokers
- Overview Forward Rates Historical Data Related Instruments Currency Converter Contracts Options

GBP/USD Forward Rates

Name	Bid	Ask	High	Low	Chg.	Time
GBPUSD ON FWD	0.0200	0.0300	0.0400	0.0270	0.0200	12:34:00
GBPUSD TN FWD	0.0090	0.0690	0.0470	0.0220	0.0020	12:35:00
GBPUSD SN FWD	0.0940	0.1230	0.1490	0.0540	0.0740	12:35:00
GBPUSD SW FWD	0.5200	0.6200	0.6800	0.5400	-0.0500	12:34:00
GBPUSD 2W FWD	0.9100	0.9600	1.0400	0.8900	-0.0800	12:35:00
GBPUSD 3W FWD	1.2100	1.4400	1.4600	1.2500	-0.1400	12:35:00
GBPUSD 1M FWD	1.8500	1.9400	2.0700	1.7900	-0.0800	12:35:00
GBPUSD 2M FWD	3.6000	3.7000	3.8300	2.7400	0.0800	12:35:00
GBPUSD 3M FWD	5.5700	5.6000	5.8000	5.4500	0.1800	12:35:00
GBPUSD 4M FWD	10.7000	10.8800	11.0600	10.4600	-0.0300	12:35:00
GBPUSD 5M FWD	11.9100	12.4600	12.5200	11.8200	-0.1700	12:34:00
GBPUSD 6M FWD	13.0200	14.1400	14.0700	13.1100	-0.1800	12:35:00
GBPUSD 7M FWD	15.2100	15.6600	16.5300	14.9200	0.0900	12:34:00
GBPUSD 8M FWD	16.2500	17.7500	17.6100	16.5700	-0.8000	12:35:00
GBPUSD 9M FWD	18.1900	19.2000	19.0000	18.3000	-0.5400	12:35:00
GBPUSD 10M FWD	19.1100	22.0500	21.6400	20.3100	-1.2900	12:35:00
GBPUSD 11M FWD	20.8300	24.0900	23.0000	22.5200	-2.0000	12:35:00
GBPUSD 1Y FWD	19.9800	29.9800	25.4100	24.5100	-3.8500	12:35:00
GBPUSD 15M FWD	30.4000	33.4000	33.4700	32.0700	1.0400	12:05:00
GBPUSD 21M FWD	45.0000	51.0000	48.5400	47.9700	1.5700	12:05:00
GBPUSD 2Y FWD	46.3400	65.1800	62.5000	56.1700	-4.2900	12:05:00
GBPUSD 3Y FWD	83.1000	90.6000	86.2000	87.6000	0.9000	12:04:00
GBPUSD 4Y FWD	120.3000	130.3000				
GBPUSD 5Y FWD	166.2000	181.2000				

GBP/USD Discussions

Account Payable (GBP): Original Position = Short the GBP

Purchase Order (A/P) makes the firm "short" the GBP

Date of Transaction: 9/30/2023
Date of Settlement: 11/29/2023
Days: 60

Account Payable: £1,000,000

Note: The firm is "short" the GBP because it benefits if the GBP depreciates against the USD

Spot Rate GBPUSD (\$/£1): \$1.2756

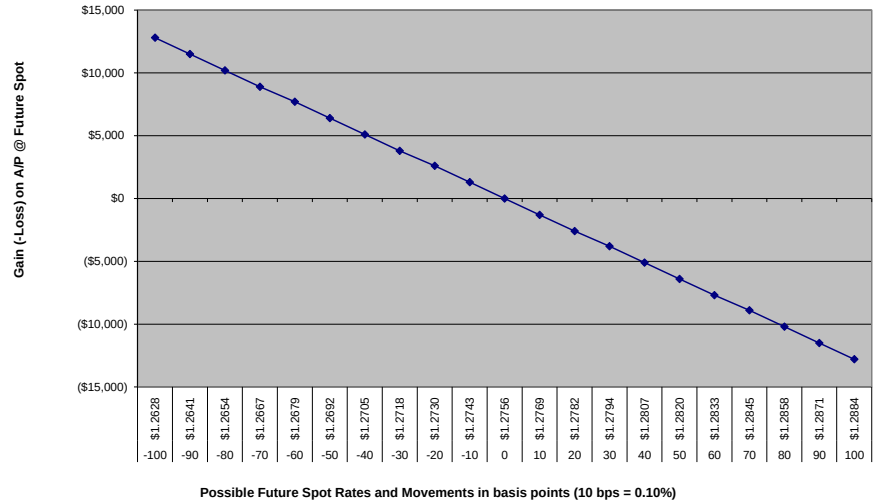
Note: The Ask price (to purchase GBP) and a direct quote = the GBP moves (up/down) with the quote (up/down)

Book value of A/P: \$1,275,600

Note: The firm is exposed insofar as the value of this A/P in USD is not certain because of the uncertain future spot rate

Movement Basis Points	Movement as a Percent	Possible Future Spot	Book Value of A/P	A/P Adjusted at Future Spot	Gain (Loss) on A/P
-100	-1.00%	\$1.2628	\$1,275,600	\$1,262,800	\$12,800
-90	-0.90%	\$1.2641	\$1,275,600	\$1,264,100	\$11,500
-80	-0.80%	\$1.2654	\$1,275,600	\$1,265,400	\$10,200
-70	-0.70%	\$1.2667	\$1,275,600	\$1,266,700	\$8,900
-60	-0.60%	\$1.2679	\$1,275,600	\$1,267,900	\$7,700
-50	-0.50%	\$1.2692	\$1,275,600	\$1,269,200	\$6,400
-40	-0.40%	\$1.2705	\$1,275,600	\$1,270,500	\$5,100
-30	-0.30%	\$1.2718	\$1,275,600	\$1,271,800	\$3,800
-20	-0.20%	\$1.2730	\$1,275,600	\$1,273,000	\$2,600
-10	-0.10%	\$1.2743	\$1,275,600	\$1,274,300	\$1,300
0	0.00%	\$1.2756	\$1,275,600	\$1,275,600	\$0
10	0.10%	\$1.2769	\$1,275,600	\$1,276,900	(\$1,300)
20	0.20%	\$1.2782	\$1,275,600	\$1,278,200	(\$2,600)
30	0.30%	\$1.2794	\$1,275,600	\$1,279,400	(\$3,800)
40	0.40%	\$1.2807	\$1,275,600	\$1,280,700	(\$5,100)
50	0.50%	\$1.2820	\$1,275,600	\$1,282,000	(\$6,400)
60	0.60%	\$1.2833	\$1,275,600	\$1,283,300	(\$7,700)
70	0.70%	\$1.2845	\$1,275,600	\$1,284,500	(\$8,900)
80	0.80%	\$1.2858	\$1,275,600	\$1,285,800	(\$10,200)
90	0.90%	\$1.2871	\$1,275,600	\$1,287,100	(\$11,500)
100	1.00%	\$1.2884	\$1,275,600	\$1,288,400	(\$12,800)

Risk profile of A/P in GBP at uncertain Future Spot Rates



Account Payable (GBP) with Forward Hedge

Purchase Order (A/P) makes the firm "short" the GBP

Date of Transaction: 9/30/2023
Date of Settlement: 11/29/2023
Days: 60

Account Payable: £1,000,000

Note: This requires a 6-month forward rate

Spot Rate GBPUSD (\$/£1): \$1.2756

Note: The firm is "short" the GBP because it benefits if the GBP depreciates against the USD

Book value of A/P: \$1,275,600

Note: This is a direct quote = the foreign currency moves (up/down) with the quote (up/down)

Bank quoted Forward Rate (pips): 3.7000

Note: The firm is exposed insofar as the final (settled) value of this A/P in USD is not certain because of the uncertain future spot rate

Bank quoted Forward Rate GBPUSD (\$/£1): \$1.2760

Note: The pip quote must be divided by 10,000, this quotient is then added to the spot rate in order to calculate the forward rate

Forward Premium/Discount: 0.03%

Note: The GBP is appreciating in the Forward Rate compared to the Spot Rate, this increases the A/P as measured in USD

Settlement value of A/P @ Forward Rate: \$1,275,970

Note: The Forward Rate is quoted as a premium or discount (expressed as a percent above/below the Spot Rate)

Net Gain(Loss) @ Forward Rate: -\$370

Note: The value of the A/P is more when converted at the Forward Rate

Gain (Loss) as % -0.03%

Note: The firm incurs a foreign currency exchange loss

Note: The foreign currency exchange loss is expressed as a percent (equal to but opposite the Forward premium calculated above)

Hypothetical Spot Movement -0.75%

Note: The foreign currency exposure remains unhedged until the A/P settles and this is the hypothetical spot movement of the GBP

Spot Rate GBPUSD (\$/£1): \$1.2660

Note: The new spot rate on date the A/P settles based on hypothetical movement

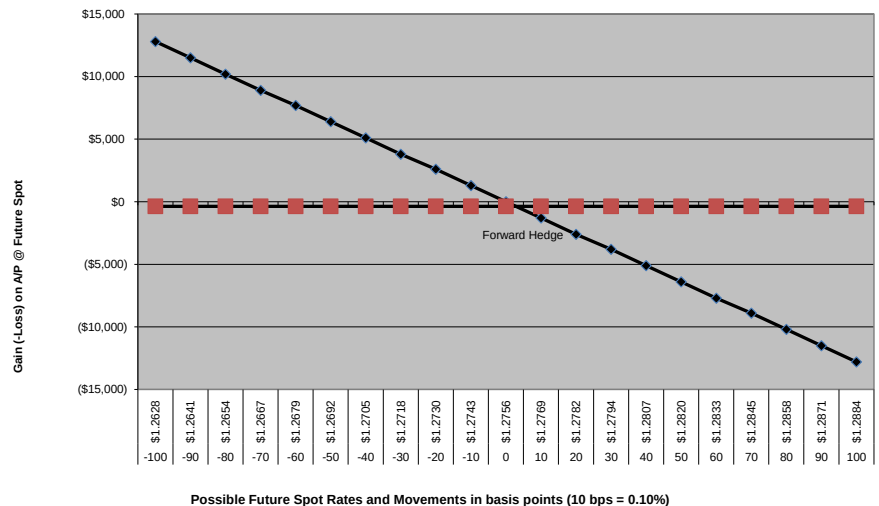
Settlement value of A/P @ Hypothetical Spot Rate: \$1,266,000

Net Gain(Loss) @ Hypothetical Spot Rate: \$9,600

Note: The foreign currency exchange loss is expressed as a percent (equal to but opposite the Forward premium calculated above)

Movement Basis Points	Movement as a Percent	Possible Future Spot	Book Value of A/P	A/P Adjusted at Future Spot	Gain (Loss) on A/P
-100	-1.00%	\$1.2628	\$1,275,600	\$1,262,800	\$12,800
-90	-0.90%	\$1.2641	\$1,275,600	\$1,264,100	\$11,500
-80	-0.80%	\$1.2654	\$1,275,600	\$1,265,400	\$10,200
-70	-0.70%	\$1.2667	\$1,275,600	\$1,266,700	\$8,900
-60	-0.60%	\$1.2679	\$1,275,600	\$1,267,900	\$7,700
-50	-0.50%	\$1.2692	\$1,275,600	\$1,269,200	\$6,400
-40	-0.40%	\$1.2705	\$1,275,600	\$1,270,500	\$5,100
-30	-0.30%	\$1.2718	\$1,275,600	\$1,271,800	\$3,800
-20	-0.20%	\$1.2730	\$1,275,600	\$1,273,000	\$2,600
-10	-0.10%	\$1.2743	\$1,275,600	\$1,274,300	\$1,300
0	0.00%	\$1.2756	\$1,275,600	\$1,275,600	\$0
10	0.10%	\$1.2769	\$1,275,600	\$1,276,900	(\$1,300)
20	0.20%	\$1.2782	\$1,275,600	\$1,278,200	(\$2,600)
30	0.30%	\$1.2794	\$1,275,600	\$1,279,400	(\$3,800)
40	0.40%	\$1.2807	\$1,275,600	\$1,280,700	(\$5,100)
50	0.50%	\$1.2820	\$1,275,600	\$1,282,000	(\$6,400)
60	0.60%	\$1.2833	\$1,275,600	\$1,283,300	(\$7,700)
70	0.70%	\$1.2845	\$1,275,600	\$1,284,500	(\$8,900)
80	0.80%	\$1.2858	\$1,275,600	\$1,285,800	(\$10,200)
90	0.90%	\$1.2871	\$1,275,600	\$1,287,100	(\$11,500)
100	1.00%	\$1.2884	\$1,275,600	\$1,288,400	(\$12,800)

Risk profile of A/P in GBP with Forward Contract Hedge



EUR/USD - Euro US Dollar

 Real-time FX

1.1688 -0.0018 (-0.15%)

09:04:40 - Real-time Data. ([Disclaimer](#))

[Add to Portfolio](#)

[Create Alert](#)

Type: Currency
Group: Major
Base: Euro
Second: US Dollar

Prev. Close: 1.1706 Bid/Ask: 1.1687 / 1.1690 Day's Range: 1.1672 - 1.1720

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[Overview](#) [Forward Rates](#) [Historical Data](#) [Related Instruments](#) [Currency Converter](#) [Contracts](#) [Options](#)

EUR/USD Forward Rates

Name	Bid	Ask	High	Low	Chg.	Time
EURUSD ON FWD	0.1990	0.2330	0.2200	0.2100	0.0390	12:31:00
EURUSD TN FWD	0.2120	0.2260	1.4000	0.0150	0.0000	12:31:00
EURUSD SN FWD	0.6500	0.6780	0.6700	0.2300	0.4800	12:33:00
EURUSD SW FWD	1.7400	1.9000	1.9300	1.8100	-0.1300	12:34:00
EURUSD 2W FWD	3.4400	3.5000	3.6900	3.4500	-0.0400	12:33:00
EURUSD 3W FWD	5.1200	5.2700	5.3700	5.1700	-0.1300	12:29:00
EURUSD 1M FWD	7.5600	7.6600	7.9700	7.4700	-0.3500	12:33:00
EURUSD 2M FWD	14.8700	15.0700	15.6400	14.3400	0.0500	12:34:00
EURUSD 3M FWD	22.9700	23.1700	23.2900	22.4300	0.7100	12:33:00
EURUSD 4M FWD	33.6600	34.1600	34.4500	33.5100	-0.5700	12:33:00
EURUSD 5M FWD	41.5200	41.7700	41.8900	40.9800	0.0000	12:34:00
EURUSD 6M FWD	48.2000	49.0000	48.9700	47.9000	-0.2600	12:33:00
EURUSD 7M FWD	56.6800	57.6800	57.4500	56.4200	-0.4400	12:33:00
EURUSD 8M FWD	63.9700	64.9700	65.4300	63.8100	-0.2700	12:33:00
EURUSD 9M FWD	71.7600	72.7600	74.0900	70.5100	-0.6700	12:34:00
EURUSD 10M FWD	80.0700	81.4200	82.1700	80.3400	-0.8000	12:33:00
EURUSD 11M FWD	88.0400	89.1900	88.9700	88.2000	-0.4900	12:33:00
EURUSD 1Y FWD	96.4200	98.8000	98.4000	96.7800	-0.7800	12:34:00
EURUSD 15M FWD	121.4600	122.9600	122.7700	121.7100	0.3900	12:04:00
EURUSD 21M FWD	172.5600	175.4400	174.1800	171.6900	-0.1400	12:03:00
EURUSD 2Y FWD	197.8300	201.1300	206.4000	199.7700	-2.3900	12:03:00
EURUSD 3Y FWD	302.7100	309.9100	308.2100	307.4200	-6.2900	12:03:00
EURUSD 4Y FWD	410.7500	450.0000				
EURUSD 5Y FWD	566.0000	576.0000				

EUR/USD Discussions

Account Receivable (EUR): Original Position = Long Euros

Sales Order (A/R) makes the firm "long" the EUR

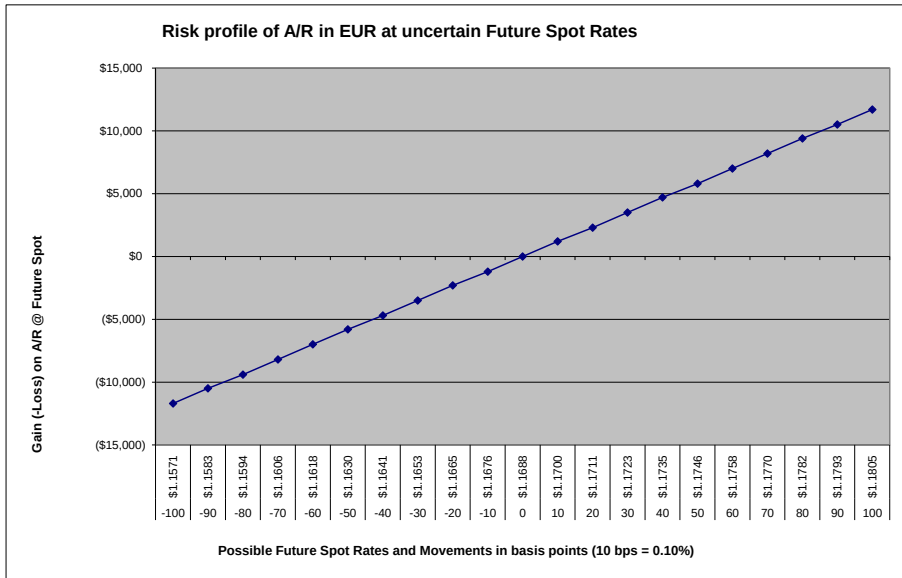
Date of Transaction: 9/30/2023
Date of Settlement: 11/29/2023
Days: 60

Account Receivable: € 1,000,000 Note: The firm is "long" the EUR because it benefits if the EUR appreciates against the USD

Spot Rate EURUSD (\$/€): **\$1.1688** Note: Bid price (to sell EUR) and a direct quote = the EUR moves (up/down) with the quote (up/down)

Book value of A/R: **\$1,168,800** Note: The firm is exposed insofar as the final (settled) value of this A/R in USD is not certain because of the uncertain future spot rate

Movement Basis Points	Movement as a Percent	Possible Future Spot	Book Value of A/R	A/R Adjusted at Future Spot	Gain (Loss) on A/R
-100	-1.00%	\$1.1571	\$1,168,800	\$1,157,100	(\$11,700)
-90	-0.90%	\$1.1583	\$1,168,800	\$1,158,300	(\$10,500)
-80	-0.80%	\$1.1594	\$1,168,800	\$1,159,400	(\$9,400)
-70	-0.70%	\$1.1606	\$1,168,800	\$1,160,600	(\$8,200)
-60	-0.60%	\$1.1618	\$1,168,800	\$1,161,800	(\$7,000)
-50	-0.50%	\$1.1630	\$1,168,800	\$1,163,000	(\$5,800)
-40	-0.40%	\$1.1641	\$1,168,800	\$1,164,100	(\$4,700)
-30	-0.30%	\$1.1653	\$1,168,800	\$1,165,300	(\$3,500)
-20	-0.20%	\$1.1665	\$1,168,800	\$1,166,500	(\$2,300)
-10	-0.10%	\$1.1676	\$1,168,800	\$1,167,600	(\$1,200)
0	0.00%	\$1.1688	\$1,168,800	\$1,168,800	\$0
10	0.10%	\$1.1700	\$1,168,800	\$1,170,000	\$1,200
20	0.20%	\$1.1711	\$1,168,800	\$1,171,100	\$2,300
30	0.30%	\$1.1723	\$1,168,800	\$1,172,300	\$3,500
40	0.40%	\$1.1735	\$1,168,800	\$1,173,500	\$4,700
50	0.50%	\$1.1746	\$1,168,800	\$1,174,600	\$5,800
60	0.60%	\$1.1758	\$1,168,800	\$1,175,800	\$7,000
70	0.70%	\$1.1770	\$1,168,800	\$1,177,000	\$8,200
80	0.80%	\$1.1782	\$1,168,800	\$1,178,200	\$9,400
90	0.90%	\$1.1793	\$1,168,800	\$1,179,300	\$10,500
100	1.00%	\$1.1805	\$1,168,800	\$1,180,500	\$11,700



Account Receivable (EUR) with Forward Hedge

Sales Order (A/R) makes the firm "long" the EUR

Date of Transaction: 9/30/2023
Date of Settlement: 11/29/2023
Days: 60

Account Receivable: € 1,000,000 Note: This requires a 9-month forward rate

Spot Rate EURUSD (\$/€): **\$1.1688** Note: The firm is "long" the EUR because it benefits if the EUR appreciates against the USD

Book value of A/R: **\$1,168,800** Note: This is a direct quote = the foreign currency moves (up/down) with the quote (up/down)

Bank quoted Forward Rate (pips): 14.8700 Note: The firm is exposed insofar as the final (settled) value of this A/R in USD is not certain because of the uncertain future spot rate

Bank quoted Forward Rate EURUSD (\$/€): **\$1.1703** Note: The pip quote must be divided by 10,000, this quotient is then added to the spot rate in order to calculate the forward rate

Forward Premium/Discount: 0.13% Note: The EUR is appreciating in the Forward Rate compared to the Spot Rate, this increases the A/R as measured in USD

Settlement value of A/R @ Forward Rate: **\$1,170,287** Note: The Forward Rate is quoted as a premium or discount (expressed as a percent above/below the Spot Rate)

Net Gain(Loss) @ Forward Rate: **\$1,487** Note: The value of the A/R is higher when converted at the Forward Rate

Gain (Loss) as % 0.13% Note: The firm incurs a foreign currency exchange gain

Note: The foreign currency exchange gain is expressed as a percent (equal to the Forward premium percent)

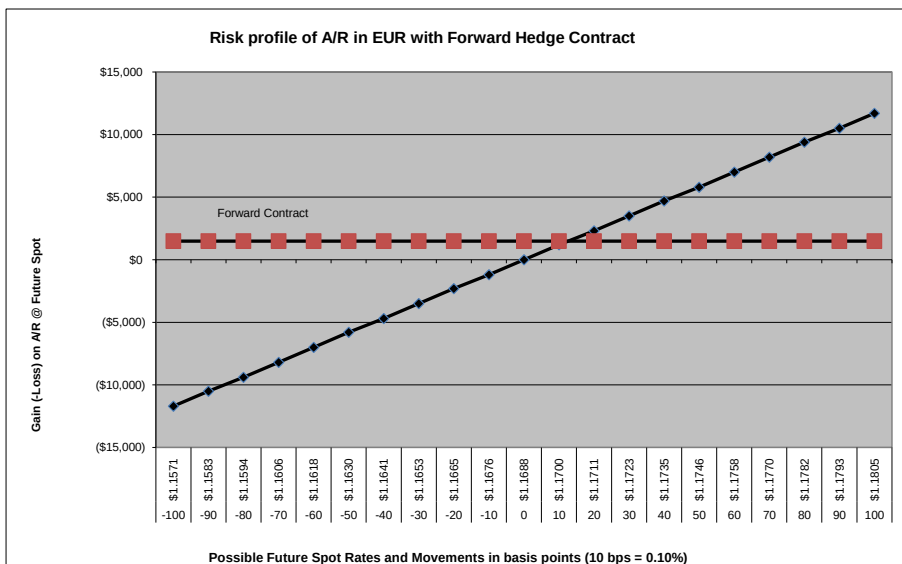
Hypothetical Spot Movement -0.50% Note: The foreign currency exposure remains unhedged until the A/P settles and this is the hypothetical spot movement of the GBP

Spot Rate GBPUSD (\$/£1): **\$1.1630** Note: The new spot rate on date the A/P settles based on hypothetical movement

Settlement value of A/P @ Hypothetical Spot Rate: **\$1,163,000**

Net Gain(Loss) @ Hypothetical Spot Rate: **-\$5,800** Note: The foreign currency exchange loss is expressed as a percent (equal to but opposite the Forward premium calculated above)

Movement Basis Points	Movement as a Percent	Possible Future Spot	Book Value of A/R	A/R Adjusted at Future Spot	Gain (Loss) on A/R
-100	-1.00%	\$1.1571	\$1,168,800	\$1,157,100	(\$11,700)
-90	-0.90%	\$1.1583	\$1,168,800	\$1,158,300	(\$10,500)
-80	-0.80%	\$1.1594	\$1,168,800	\$1,159,400	(\$9,400)
-70	-0.70%	\$1.1606	\$1,168,800	\$1,160,600	(\$8,200)
-60	-0.60%	\$1.1618	\$1,168,800	\$1,161,800	(\$7,000)
-50	-0.50%	\$1.1630	\$1,168,800	\$1,163,000	(\$5,800)
-40	-0.40%	\$1.1641	\$1,168,800	\$1,164,100	(\$4,700)
-30	-0.30%	\$1.1653	\$1,168,800	\$1,165,300	(\$3,500)
-20	-0.20%	\$1.1665	\$1,168,800	\$1,166,500	(\$2,300)
-10	-0.10%	\$1.1676	\$1,168,800	\$1,167,600	(\$1,200)
0	0.00%	\$1.1688	\$1,168,800	\$1,168,800	\$0
10	0.10%	\$1.1700	\$1,168,800	\$1,170,000	\$1,200
20	0.20%	\$1.1711	\$1,168,800	\$1,171,100	\$2,300
30	0.30%	\$1.1723	\$1,168,800	\$1,172,300	\$3,500
40	0.40%	\$1.1735	\$1,168,800	\$1,173,500	\$4,700
50	0.50%	\$1.1746	\$1,168,800	\$1,174,600	\$5,800
60	0.60%	\$1.1758	\$1,168,800	\$1,175,800	\$7,000
70	0.70%	\$1.1770	\$1,168,800	\$1,177,000	\$8,200
80	0.80%	\$1.1782	\$1,168,800	\$1,178,200	\$9,400
90	0.90%	\$1.1793	\$1,168,800	\$1,179,300	\$10,500
100	1.00%	\$1.1805	\$1,168,800	\$1,180,500	\$11,700



Topic 5: Currency Derivatives

Hedging Foreign Currency Exchange Risk

As we have learned, the two types of foreign currency transaction risk faced by firms are:

1. **Receivable (A/R).** A firm that will receive a future payment in a foreign currency must use the foreign currency it receives (e.g. £) on the settlement date to purchase its home currency (e.g. \$). This firm faces the risk that the value of the foreign currency (as expressed in the exchange rate) will decrease and thereby purchase fewer dollars, thus decreasing the amount received when it is converted into its home currency. *The firm will suffer a loss if the foreign currency depreciates against the home currency, and will gain if the foreign currency appreciates against the home currency.* The firm is “long” the foreign currency.
2. **Payable (A/P).** A firm that is required to make a future payment in a foreign currency must use its home currency (e.g. \$) to purchase the foreign currency (e.g. ¥) in order to meet the payable on the settlement date. This firm faces the risk that the price of the foreign currency (as expressed in the exchange rate) will increase, thus increasing the amount of home currency is required to cover the payment in the foreign currency. *The firm will suffer a loss if the foreign currency appreciates against the home currency, and it will gain if the foreign currency depreciates against the home currency.* The firm is “short” the foreign currency.

In summary, foreign currency exchange risk is created by the **uncertainty of future spot rates**.

Consider the example of an A/R that is due in 90 days. When the transaction involves foreign currency, your firm “books” the sale but the customer (on the date of settlement) will make payment in a foreign currency at some future date. Three facts are relevant:

1. The A/R is “booked” by your firm on the date the transaction (sales order) occurs, and
2. The “book value” of the A/R is determined by converting the foreign currency into the home currency at the spot rate on the transaction date, and
3. Without taking any further action, the value (in the home currency) of the foreign currency received on the date of settlement will be determined by the spot rate on the date of settlement.

Example 1: A sales order is booked today and will settle in 90 days (an A/R is created)

On 9/1/18, a European customer agrees to pay your firm **€1,000,000** for your product – payment is due on 12/1/18.

The spot rate on 9/1/18 is \$1.1461 (this is a direct quote).

Thus, your firm books the A/R on 9/1/18 as **\$1,146,100** ($€1,000,000 \times \1.1461)

On 12/1/18, the settlement date, the spot rate of is \$1.1375.

Thus, your firm receives €1,000,000 on 12/1/18 and converts it into **\$1,137,500** ($€1,000,000 \times \1.1375).

Thus, your firm incurs a “loss” on the foreign currency exchange of **\$8,600**, which is the difference between the amount booked (\$1,146,100) and the amount received (\$1,137,500)

The risk your firm faces is created by the uncertainty of what the spot rate will be in 90 days (on the settlement date) when it receives the €1,000,000. If the € depreciates between 9/1/18 and 12/1/18, your firm will have lost value in its receivable because of a change in the exchange rate that was used to calculate the book value of the A/R and the exchange rate that is used to calculate the settled value of the A/R.

We can also consider an A/P that is due on 90 days. Again, with this transaction:

1. The A/P is “booked” by your firm on the date the transaction (purchase order) occurs, and
2. The “book value” of the A/P is determined by converting the foreign currency into the home currency at the spot rate on the transaction date, and
4. Without taking any further action, the cost (in the home currency) of the foreign currency needed on the date of settlement will be determined by the spot rate on the date of settlement.

Example 2: A purchase order is booked today and will settle in 90 days (an A/P is created)

On 9/1/18, your firm agrees to pay its European supplier **€1,000,000** for its product – payment is due on 12/1/18.

The spot rate on 9/1/18 is \$1.1461.

Thus, your firm books the A/P on 9/1/18 as **\$1,146,100** ($€1,000,000 \times \1.1461)

On 12/1/18, the settlement date, the spot rate is \$1.1575.

Thus, your firm must use **\$1,157,500** to purchase €1,000,000 on 12/1/18 and make the payment ($€1,000,000 \times \$1.1575$).

Your firm incurs a “loss” on the foreign currency exchange of **\$11,400** (which is the difference between the amount booked, \$1,146,000, and the amount it costs to settle the payment, \$1,157,500)

The risk your firm faces is created by the uncertainty of what the spot rate will be in 90 days (on the settlement date) when it must pay €1,000,000. If the € appreciates between 9/1/18 and 12/1/18, the €1,000,000 will cost more in \$. Your firm will incur a loss on the cost of settling its payable because of a change in the exchange rate from the book date to the settlement date.

As the two examples above illustrate, a change in the exchange rate can result in a loss to firms that must either collect (A/R) or make future payments (A/P) in a foreign currency. Losses will occur if the foreign currency depreciates (with an A/R) or appreciates (with an A/P). Of course, the changes could result in a gain if the foreign currency appreciates (with an A/R) or depreciates (with an A/P). However, the **uncertainty** of the future spot rate represents **risk** to the firm transacting in foreign currencies.

A firm facing this risk (whether in the form of a receivable or payable) has the ability to “**hedge**” their position and reduce or eliminate the risk of an uncertain future spot rate. A “hedge” is a financial position taken to protect a firm from exposure to exchange rate fluctuations. A hedge can replace the uncertainty of the future spot rate with the certainty of a fixed future exchange rate. This “certainty” can result in either a certain gain or a certain loss. A hedge can also replace the uncertainty of future gains or losses with the certainty of a limited loss but the opportunity of an unlimited gain (see below).

Of course, the firm can choose NOT to hedge its risk exposure at all (a position that would be referred to as “unhedged” or “fully exposed”), or to only hedge a certain percentage of its total risk exposure (a position that would be referred to as “partially hedged” or “partially exposed”). A firm that remains unhedged or partially hedged is accepting the risk that a change in the exchange rate might result in a loss (a higher payable or a lower receivable). Perhaps the firm is hoping that a change in the exchange rate will result in a gain (a lower payable or a higher receivable). Nonetheless, most firms prefer to concentrate on their core business (selling products or services), thus making it a policy to “hedge” its foreign exchange rate risk as much as possible.

There are three financial instruments available to firms looking to hedge a foreign currency position:

1. Forward contracts
2. Futures contracts
3. Option contracts

Each of these hedge instruments shares similarities with the others, but each also possesses its own unique characteristics. They will be discussed below in order.

1. Foreign Currency Forward Contract

A forward contract is an agreement between a commercial bank and a firm regarding an exchange of:

1. a specific amount of currency,
2. on a specific future date (called the “settlement date”),
3. at a specific exchange rate (called the “forward rate”).

The forward contract allows a firm to “lock in” the exchange rate on the future settlement date. An exchange rate quoted today by a bank for a specific future date is called the **forward rate**. Thus, the forward rate is the exchange rate that will be used on settlement date, regardless of what the actual spot rate is on that settlement date. In other words, the uncertainty of the future spot rate has been replaced by the certainty of the forward rate.

Example 3: Using a forward contract to hedge an A/R

On 9/1/18, a European customer agrees to pay your firm **€1,000,000** for your product – payment is due on 12/1/18.

The spot rate of €/ \$ on 9/1/18 is \$1.1461.

Thus, your firm books the A/R on 9/1/18 as **\$1,146,100** ($€1,000,000 \times \1.1461)

The 90-day forward rate quoted by the bank today is \$1.1455.

Your firm agrees to a forward contract: the settlement date is 12/1/18 and the forward exchange rate is \$1.1455.

On the settlement date, your firm will receive €1,000,000 and exchange it for **\$1,145,500** with the bank ($€1,000,000 \times \$1.1455$)

Thus, your firm is certain of the amount of \$ that the € will be worth in 90 days when it receives payment in €.

Note: Your firm, by using the forward contract, has guaranteed a loss of **\$600** on the A/R, which is the difference between the amount booked (\$1,146,100) and the amount received from the bank (\$1,145,500) on the settlement date. The loss is the result of the forward rate indicating that the € is depreciating against the \$. This is acceptable to most firms because the uncertainty of the future spot rate has been replaced by the certainty of the forward rate with the bank, despite the loss.

Note: In this example, the forward rate guarantees your firm a loss on the A/R. It can also be the case that the forward rate guarantees your firm a “gain” on the A/R. That would be the case if the forward rate indicates that the € is appreciating against the \$ rather than depreciating – thus, the amount of \$ received on the settlement date would increase.

An example of using a forward contract to hedge an A/P would be identical to the A/R in Example 3 above, except that on the settlement date your firm would be exchanging dollars for the foreign currency in order to make a payment in the foreign currency. The forward rate would guarantee your firm a certain cost (in dollars) of meeting the payment due (in foreign currency) on the A/P.

Many large Multinational Corporations (MNCs) use forward contracts. Because forward contracts are used mostly by large MNCs, a typical forward transaction is valued at \$1 million or more and is not commonly used by consumers or small firms.

2. Foreign Currency Futures Contract

A foreign currency futures contract involves two parties (a “buyer” and a “seller”) agreeing to exchange a **specified quantity** of a particular currency at a **specified price** on a **specified settlement date** in the future. It is known as a “derivative” instrument (also called a contingent claim security) because its value is derived from, or contingent upon, the value of the underlying security (in this case, the underlying foreign currency). Futures contracts are commonly used by **firms** to hedge their foreign currency positions and by **speculators** hoping to capitalize on expected exchange rate movements.

In a futures contract, there are two parties: the **buyer** and the **seller** of the contract, who are also the respective “buyer” and “seller” of the underlying currency. A broker may act as an intermediary, much like transactions involving stocks and bonds, but the broker is not the buyer or the seller of the contract itself. The futures contract represents an obligation for both parties to exchange the currency (i.e. settle the contract) on a specific future date. The buyer buys the currency that the seller sells.

Every futures contract has three characteristics:

1. **Number of currency units** (amount of foreign currency per contract).
2. **Price** (exchange rate of the foreign currency in USD on the settlement date).
3. **Settlement date** (future date on which the contract “settles” and the currency is exchanged)

The **buyer** of a currency futures contract agrees to buy the foreign currency (i.e. he agrees to pay a fixed amount of US dollars for the foreign currency on the settlement date). The contract allows the buyer to “lock in” the exchange rate he will **pay** for the foreign currency on the settlement date. (Note: the buyer of a futures contract would be the party needing to purchase the foreign currency in order to make a future payment – **it is the hedge position for a payable.**)

The **seller** of a futures contract agrees to sell the foreign currency (i.e. he agrees to accept a fixed amount of US dollars for the foreign currency on the settlement date). The contract allows the seller to “lock in” the exchange rate he will receive for the foreign currency on the settlement date. (Note: the seller of a futures contract would lock in the amount of dollars received when exchanging foreign currency received in the future – **it is the hedge position for a receivable.**)

Currency futures contracts are available for many of the most widely traded currencies at the Chicago Mercantile Exchange (CME) and through the NASDAQ OMX Futures Exchange (NFX), and the contract for each currency specifies a standardized number of units. Some examples below:

<u>Currency</u>	<u>Units per Contract – CME</u>	<u>Units per Contract - NFX</u>
Australian dollar (AUD)	100,000	10,000
Brazilian real (BRL)	100,000	10,000
British pound (GBP)	62,500	10,000
Canadian dollar (CAD)	100,000	10,000
Euro (EUR)	125,000	10,000
Japanese yen (JPY)	12,500,000	1,000,000
Mexican peso (MXN)	500,000	100,000
New Zealand dollar (NZD)	100,000	10,000
Russian ruble (RUB)	2,500,000	1,000,000
South African rand (SK)	500,000	100,000
Swiss franc (CHF)	125,000	10,000

Firms or individuals can execute orders (to either buy or sell a futures contract) by calling brokerage firms that serve as intermediaries. The order to buy or sell a currency futures contract (for a specific currency and at the specific settlement date) is communicated to the brokerage firm, which in turn communicates the order to the CME. A floor broker at the CME who specializes in that type of currency futures contract attempts to find a counterparty to fill the order. For example, if a firm wants to purchase a Mexican peso futures contract with a December settlement date, the broker assigned to execute this order will look for another broker who has an order to sell a Mexican peso futures contract with a December settlement date. (Note: full service brokers typically charge a flat commission for acting as the intermediary – for example, \$20 per contract.)

Currency futures contracts on the CME expire (settle) the third Wednesday in March, June, September, and December (these are the settlement dates).

Example:

Assume that on February 10, a futures contract on £62,500 (GBP) with a March settlement date is priced at \$1.50 per pound. Consider the positions of two different firms on the opposite sides of this contract.

The **buyer** of this currency futures contract will receive £62,500 on the March settlement date and will pay \$93,750 for the pounds ($£62,500 \times \$1.50 = \$93,750$).

The **seller** of this contract is obligated to sell £62,500 at a price of \$1.50 per pound and therefore will receive \$93,750 on the settlement date.

Pricing Currency Futures Contracts

The price (exchange rate) of currency futures contracts will be similar to the forward rate for a given currency at the same settlement date. This relationship is enforced by the potential arbitrage activity that would occur if significant discrepancies existed. The currency futures price differs from the spot rate for the same reason that a forward rate differs from the spot rate. (Note: the possibility of Interest Rate Arbitrage exists with a futures contract in the same way as it does with a forward contract.)

Credit Risk of a Currency Futures Contract

Each currency futures contract represents an agreement between a client and the exchange clearing house, even though the Exchange has not taken a position. To illustrate, assume you call a broker to request the purchase of a Euro futures contract with a March settlement date. Meanwhile, another person unrelated to you calls a broker to request the sale of a similar Euro futures contract. Neither party needs to worry about the credit risk of the counterparty. The exchange clearinghouse assures that you will receive whatever is owed to you as a result of the futures position.

Closing Out a Futures Position

If a firm holding a currency futures contract decides before the settlement date that it no longer wants to maintain its position, it can close out the position by selling an identical futures contract. Conversely, a firm that had sold a futures contract can close out the position by purchasing an identical contract. The gain or loss to the firm from its previous position is dependent on the price of the new position. See the example below:

Example: **Closing out a Futures Contract position**

January 10.....February 15.....March 19 (settlement date)

Step 1: Contract to Buy
Australian dollars @ \$0.53
Size: A\$100,000

Step 2: Contract to Sell
Australian dollars @ \$0.50
Size: A\$100,000

Step 3: Settle Contracts
- \$53,000 = Payment Contract #1
+ \$50,000 = Receipt Contract #2

\$53,000 at settlement date

\$50,000 at settlement date

- \$3,000 = loss at settlement date

Sellers of a futures contract in Step #1 can close out their position by purchasing a futures contract of similar size and with similar settlement date in Step #2.

The price of a futures contract changes over time in accordance with movements of the spot rate, and also with changing expectations about the currency's value as of the settlement date (the forward rate).

How Firms Use Currency Futures to Hedge Foreign Currency Risk

The purchase of a futures contract locks in the price at which a firm can purchase a currency. This strategy would be used to hedge a payable.

Example:

A firm orders Canadian goods and upon delivery will need to send C\$500,000 to the Canadian exporter in June. The firm thus purchases five Canadian dollar futures contracts today with a June settlement date. The price of the futures contract locks in the price to be paid for Canadian dollars in June. Why do they purchase five contracts?

The sale of a futures contract locks in the price at which a firm can sell currency. This strategy would be used to hedge a receivable.

Example:

An exporter to Great Britain will receive £125,000 in September for a shipment of goods to London. The firm thus sells two British pound futures contract today with a September settlement date. The price of the futures contract locks in the received pounds in September. Why do they purchase two contracts?

3. Foreign Currency Options

A foreign currency option is a contract giving the owner the **right** (but *not the obligation*) to buy or sell a **specified quantity** of foreign currency at a **specified price** prior to a or on a **specified date** in the future. It is a derivative (also called a contingent claim security) because its value is derived from, or contingent upon, the value of the underlying security (in this case, the underlying foreign currency).

In every option contract, there are two parties: the **buyer** (also called the “owner”) and the **seller** (also called the “writer”) of the **option contract itself**. In this sense, the terms “buyer” and “seller” do not refer to the buying or selling of currency, but to the buying or selling of the option contract. The buying or selling of the currency is determined by the type of option contract: a **call option** or a **put option** (discussed below). The **right to exercise the contract is held by the buyer/owner**. The seller/writer has an **obligation to perform** if the buyer/owner executes.

Option contracts that trade over-the-counter (OTC) can be tailor-made for both size and expiration date. Option contracts that trade on exchanges (such as the CME) are standardized for size and expiration date. For example, British Pound options on the CME are standardized at £62,500 per contract and expire on the Friday preceding the third Wednesday of each quarterly month (March, June, September, and January).

Two types of foreign currency option contracts are available: Calls and Puts.

Call Option: A foreign currency call option gives the buyer (owner) the right to *purchase* the foreign currency from the seller (writer) of the option at a pre-determined price (exchange rate). This is similar to being the “buyer” of a futures contract. The owner of a call option is “long” the foreign currency.

Put Option: A foreign currency put option gives the buyer (owner) the right to *sell* the foreign currency to the seller (writer) of the option at a pre-determined price (exchange rate). The owner of the put option is said to be “short” the foreign currency. This is similar to being the “seller” of a futures contract. The owner of a put option is “short” the foreign currency.

There are three critical components of every currency option (Strike Price, Premium, Expiration Date):

1. **Strike Price (Exercise Price):** The strike price (also called the exercise price) is the designated price (or exchange rate) that the owner of the option contract is allowed to purchase (call) or sell (put) the foreign currency. For example, a British Pound (£) option contract trading on the CME with a strike price of \$1.6500 allows the owner of the contract to either buy (if it is a call option) or sell (if it is a put option) £62,500 for \$103,125 (at the exchange rate of £1 per \$1.6500 or £62,500 x \$1.6500 = \$103,125).
 - a. **Call Option:** The strike price represents the exchange rate at which the owner of the option can *purchase* the foreign currency from the writer of the option. Hence, when the strike price is *lower than the spot rate*, the owner of the call option can *purchase* the foreign currency from the writer of the call option at lower strike price, then immediately sell it at the higher spot rate, making a profit. The call option contract is said to be “**in the money**” when the spot rate is higher than the strike price. Consequently, *the owner of a call option will only exercise the contract if the spot rate exceeds the strike price prior to the expiration of the contract*. The owner of a call is said to be “long” the foreign currency because he wants it to appreciate above the strike price.
 - b. **Put Option:** The strike price represents the exchange rate at which the owner of the option can *sell* the foreign currency to the writer of the option. Hence, when the strike price is *higher than the spot rate*, the owner of the put option can purchase the foreign currency at the lower spot rate, then immediately *sell* it to the writer of the put option at the higher strike price,

making a profit. The put option contract is said to be “*in the money*” when the spot is lower than the strike price. Consequently, *the owner of a put option will only exercise the contract if the spot rate is lower than the strike price prior to the expiration of the contract.* The owner of a put is said to be “short” the foreign currency because he wants it to depreciate below the strike price.

2. **Premium:** The premium is the “price” of the option contract itself (paid by the buyer of the option to the seller of the option) and is quoted in a term that is identical to the strike price. For example, in the option contract on British Pounds with a strike price of \$1.6500, the premium might be quoted as \$0.0125. Thus, the buyer of the option must pay the seller of the option \$0.0125 for every £1 that is covered by the option contract. *The option premium represents the maximum loss to the owner of the option contract because the option contract is a right and not an obligation, and the owner of the option is not required to exercise the contract if it is not “in the money.”* In our example, the \$0.0125 premium represents a cost to the buyer of the option of \$781.25 per contract (£62,500 x \$0.0125).
 - a. **Call Option:** With a call option, *the strike price plus (+) the premium* represents the “break even” point for the owner of the option. In our example of the option on British Pounds, if the strike price is \$1.6500 and the premium is \$0.0125, then the break-even point for the owner of the call option is \$1.6625. *The owner of the option will exercise the contract if the spot rate exceeds the strike price prior to expiration, but the owner will only profit from the contract if the spot rate exceeds the strike price + the premium.* If the call option is exercised on a day when the spot rate exceeds the strike price but is still less than the strike price + the premium, then the owner of the call option will lose money on the contract (but the loss will be less than the premium paid to purchase the contract).
 - b. **Put Option:** With a put option, the strike price minus (–) the premium represents the “break even” point for the owner of the option. In our example of the option on British Pounds, if the strike price is \$1.6500 and the premium is \$0.0125, then the break-even point for the owner of the put option is \$1.6375. *The owner of the option will exercise the contract if the spot rate is lower than the strike price prior to expiration, but the owner will only profit from the contract if the spot rate is lower than the strike price – the premium.* If the put option is exercised on a day when the spot rate is lower than the strike price, but is not lower than the strike price – the premium, then the owner of the put option will lose money on the contract (but the loss will be less than the premium paid to purchase the contract).
3. **Expiration Date:** The expiration date on the option contract represents the last day that the option contract may be exercised by the owner. On exchanges such as the CME, all options that are in the money are automatically exercised on the expiration date.

Note: On the CME, the British Pound option contract used as our example in this memo would be listed at 1650 (representing a strike price of \$1.6500 per £1) and the premium would be quoted under the daily settlement column (SETT) as 1.25 (representing a premium of \$0.0125 per £1).

A Euro option contract on the CME is standardized at €125,000 per contract and would be listed similarly to the British Pound (for example, a Euro option listed at 1075 would represent a strike price of \$1.0750 per €1), while the premium on the Euro option contract would be quoted under the daily settlement column (SETT) as 0030 would represent \$0.0030 per €1). Thus quoted, the “break-even” point on a Euro call option would be \$1.0780 while the “break-even” on a Euro put option would be \$1.0720.

Addendum to Hedging Foreign Currency Exchange Risk

Comparing the Forward, Futures, and Option Contracts

1. The three main alternatives for hedging foreign currency exchange risk are known as derivatives. Each of these derivatives are contracts that require counterparties:
 - a. Forward Contract
 - b. Futures Contract
 - c. Option Contract
2. Of these three contracts, the Forward Contract is not really a derivative instrument strictly speaking. The Forward Contract is a conventional two-party contract in which one party is buying, and the other party is selling, a good (in this case foreign currency) at a price that is set by the terms of the contract. As with conventional contracts, both parties in a Forward Contract intend to complete the transaction on the date of settlement (often called the “closing date” in contract law). Thus, at closing there is an actual exchange of one good (the foreign currency) for a payment that is based on the contract price. As a general rule, this exchange of the underlying good DOES NOT take place between the two parties of a derivative contract.
3. The Futures Contract and Option Contract also have two parties, one that is buying and one that is selling a good (in this case foreign currency). However, Futures and Options are derivative instruments because the contract itself has an inherent value that is “derived” from the value of the underlying asset on which the contract is based. This is what gives the term “derivative” to the contract (i.e., its value is “derived” from an underlying asset such as a foreign currency, or a commodity, or a share of stock, etc.). In addition, in virtually all derivative contracts the two parties do not intend to exchange the underlying good upon closing of the contract. Instead, the two parties close out their respective positions based on the inherent value of the contract. One party incurs a gain and the other party incurs a loss (in theory the two parties can “break even” if the price of the underlying asset in the contract is identical to the price in the spot market on the date of closing).

In the case of foreign currency derivative, the value of the derivative contract is calculated by the difference between the price of the underlying asset (foreign currency) that is fixed in the derivative contract and the price for that same underlying asset in the spot market. This is where the term “mark-to-market” comes from, as the process of “marking-to-market” involves calculating the value of a derivative contract based on the actual price in the spot market each day.

- a. In the case of a Futures Contract, the buyer is “in the money” if the derivative price (the settlement price) of the foreign currency is lower than the price in the spot market. This is because he has contracted to “buy” foreign currency at a settlement price that is lower than its actual value. The seller of this contract is “out of the money” because he has contracted to sell the foreign currency at a price lower than the price in the spot market. This is what gives the term “zero sum” to a derivative contract: one party gains and the other party loses, the net sum of the gain and loss is zero.
- b. Like a Futures Contract in which there is always one buyer and one seller on opposite sides of the contract, the Options contract also has two parties. However, unlike a Futures Contract in which there is only one kind of contract (one party is the buyer and the other is the seller of the “good” underlying the contract), there are two different kinds of Option Contracts: one in which the purchaser/owner of the contract agrees to buy the underlying asset (this is a Call Option), and one in which the purchaser/owner of the contract agrees to sell the underlying asset (this is called a Put Option). In both cases, the other party (the seller/writer of the contract) agrees to be the counterparty (i.e., the seller of the underlying

asset in a Call Option and the buyer of the underlying asset in a Put Option). The major difference in the Option Contract is that the purchaser/owner of the contract is not obligated to execute or complete the transaction, but has the “option” of executing the transaction (called “exercising” the option). Thus, he will only exercise the option contract if he is “in the money” and will not exercise if he is “out of the money” (and in fact the option will eventually expire unexercised if it remains out of the money). If the option expires without being exercised, the seller/writer of the option retains his fee (called the premium), and that fee represents the cost of the transaction to the buyer/owner of the option.

4. The difference between the Forward Contract (which is not a derivative strictly speaking) and the Futures and Option Contracts (which are derivatives) is an important distinction when considering a foreign currency hedge position. In any example of foreign currency transaction risk, the company has an underlying exposure which we will call the Original Position. In transaction risk, the Original Position is always either a Receivable or a Payable. Given this Original Position and the need to manage the exposure, the three contract hedge positions can be described as follows:
 - a. Forward Contract Hedge – this hedge simply involves an agreement to buy or sell the foreign currency contained in the Original Position at some fixed future date. Once the firm agrees to a forward contract, the uncertainty of closing out the Original Position is eliminated. The Original Position (the Receivable or Payable) remains on the books until it is closed out (settled), but the uncertainty of its “settlement value” has been eliminated. The firm waits for the date of settlement, exchanges the foreign currency at the price in the Forward Contract, and closes out the Original Position with a gain, loss, or breakeven (which depends on the difference between the spot rate on the date the transaction was booked and the forward rate).
 - b. Futures Contract Hedge – this hedge involves the firm taking a position in the derivative instrument (the Futures Contract) that is opposite its Original Position. In other words, if the Original Position is a Receivable (long), the firm takes the position of the Seller (short) in the Futures Contract. If the Original Position is a Payable (short), the firm takes the position of the Buyer (long) in the Futures Contract. As with the Forward Contract, the Original Position remains on the books until it is closed out (settled), but the uncertainty of its settlement value HAS NOT been eliminated. Instead, the firm has taken another position (in the Futures hedge) that will gain as the Original Position loses, but will also lose as the Original Position gains. Thus, the “net value” of the two positions is guaranteed, which is what eliminates the uncertainty for the firm. At some future date, the Original Position closes out (settles) at some gain or loss depending the difference between the spot rate on the date the transaction was booked and the spot rate on the date the transaction settles. Similarly, at some future date, the Futures Contract closes out (settles) at some gain or loss depending on the difference between the price in the Futures Contract (called the settlement price) and the spot rate on the date of settlement. The two positions produce a “net” result that represents the overall gain or loss to the firm.
 - c. Option Contract Hedge – this hedge also involves the firm taking a position in the derivative instrument (the Option Contract) that is opposite its Original Position. In other words, if the Original Position is a Receivable (long), the firm purchases a Put (short). If the Original Position is a Payable (short), the firm purchases a Call (long). All conditions are the same as with the Futures Contract described above, with one big difference: the “loss” in the Option hedge is limited to the cost of the Option Contract (the premium). Thus, the firm will benefit if the Original Position gains beyond the cost of the Option Contract because at that point there is no longer an offsetting loss, as is the case with a Futures Contract. However, the firm will guarantee its maximum loss because as the Original Position loses, the Option Contract gains, thus the two positions produce a “net” result.

Summary of Hedging Alternatives

	A/R	A/P
Original FOREX Position	Long	Short
Why?	Gain if FOREX ↑ FOREX ↓	Gain of FOREX ↓ FOREX ↑
Insure Against		
Forward Contract	Sell FOREX for USD	Buy FOREX w/USD
Futures Contract	Sell	Buy
Option Contract	Purchase a PUT	Purchase a CALL

	Forward Contract	Futures Contract	Option Contract
Market for Counterparty	Private (w/Bank)	Exchange or OTC	Exchange or OTC
Amount of FOREX	Customized	Standardized	Standardized
Contract Termination	"Execution Date"	"Settlement Date"	"Expiration Date"
Date of Contract Termination	Customized	Standardized	Standardized
FOREX Exchange Basis	"Forward Rate"	"Settlement Price"	"Strike Price"
Cost	Fee (Bank)	Fee (Exchange/Broker)	Premium (Writer) + Fee
Hedge Position	Certain (Fixed)	Certain (Fixed)	Certain Floor
Net Position	Certain (Fixed)	Certain (Fixed)	Certain Floor
Delivery of FOREX	Full Performance	Net Position Only	Net Position Only
Risk	Counterparty Default	Counterparty Default	Counterparty Default

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In-Class Case on Foreign Currency Transaction Exposure

Forward, Futures, Options

In-Class Case: RECEIVABLE

Your company received an order from a retailer in Germany for €2,000,000 of your product, with delivery in 90 days and the order paid in EUR. You are concerned that the EUR will depreciate versus the USD over the next three months, thus reducing profits when the A/R (in EUR) is collected and converted to USD.

You decide to consider three alternatives for hedging the exchange rate exposure:

- Forward Contract with your bank
- Futures Contract on the OTC market
- Options Contract on the Chicago Mercantile Exchange

Since your company is “long” the EUR in the A/R, you know the hedging position must be “short” the EUR. Thus, you will be the “seller” of the EUR in the Forward and Futures Contracts, and you will need to purchase a Put Option Contract.

Your bank quotes a EURUSD spot rate of \$1.1500 (direct quote BID) and a three-month EURUSD forward rate of 175.0000 (BID). The banker offers to set up a forward contract for exchanging the EUR received by your company in 90 days.

You then call a FOREX broker who quote a December EUR Futures Sell contract priced at \$1.1650 (not including fees) and a December EUR Put Option with a strike price of \$1.1650 and a premium of \$0.0200.

Complete the following:

1. Calculate the book value of the A/R on the transaction date (the date it is booked).
2. Calculate and graph the risk exposure of the A/R.
3. Calculate the forward premium/discount of the EUR based on the forward rate.
4. Calculate the gain/loss on the A/R based on the forward rate and graph the forward hedge line.
5. Calculate the premium/discount of the EUR based on the price in the Futures Contract
6. Calculate and graph the risk profile of the EUR Futures Contract.
7. Calculate and graph the risk profile of the “net” position of the A/R and the Futures contract, and the gain/loss on the “net position”
8. Calculate and graph the risk profile of the EUR Put Option, including the total cost of the option hedge, the point at which the option is “in the money” and the break-even point of the option.
9. Calculate and graph the risk profile of the “net” position of the A/R and the Put Option, including the “floor” of the net position.

Discuss the controller’s alternatives and make a recommendation.

In-Class Case: PAYABLE

Your company placed an order with a supplier in England for £2,000,000 of product with delivery in 90 days and the order paid in GBP. You are concerned that the GBP will appreciate versus the USD over the next three months, thus increasing costs when the GBP payable is settled.

You decide to consider three alternatives for hedging the exchange rate exposure:

- Forward Contract with your bank
- Futures Contract on the OTC market
- Options Contract on the Chicago Mercantile Exchange

Since you are “short” the GBP in the A/P, you know the hedging position must be “long” the GBP. Thus, you will be the “buyer” of the GBP in the Forward and Futures contracts, and you will need to purchase a Call Option Contract.

Your bank quotes a GBPUSD spot rate of \$1.3000 (direct quote ASK) and a three-month GBPUSD forward rate of 185.0000 (ASK). The banker offers to set up a forward contract for exchanging the GBP received by your company in 90 days.

You then call a FOREX broker who quote a December GBP Futures Sell contract priced at \$1.3150 (not including fees) and a December GBP Put Option with a strike price of \$1.3150 and a premium of \$0.0050.

1. Calculate the book value of the A/P on the transaction date (the date it is booked).
2. Calculate and graph the risk exposure of the A/P.
3. Calculate the forward premium/discount of the GBP based on the forward rate.
4. Calculate the gain/loss on the A/P based on the forward rate, and graph the forward hedge line.
5. Calculate the premium/discount of the GBP based on the price in the Futures Contract
6. Calculate and graph the risk profile of the GBP Futures Contract.
7. Calculate and graph the risk profile of the “net” position of the A/P and the Futures contract, and the gain/loss on the “net position”
8. Calculate and graph the risk profile of the GBP Call Option, including the total cost of the option hedge, the point at which the option is “in the money” and the break-even point of the option.
9. Calculate and graph the risk profile of the “net” position of the A/P and the Call Option, including the “floor” of the net position.

Discuss the controller’s alternatives and make a recommendation.

International Finance

In-Class Case on Transaction Exposure - Summary of Hedge Positions

Original Transaction

Transaction Date
Transaction Settlement Date
Transaction Term
Transaction Amount
Spot Rate
Transaction Book Value
Original Position in FOREX

Account Receivable

September 30, 2024
December 29, 2024
90 days (3 months)
€2,000,000
\$1.1500
\$2,300,000
Long

Account Payable

September 30, 2024
December 29, 2024
90 days (3 months)
£2,000,000
\$1.3000
\$2,600,000
Short

Forward Contract Hedge

Forward Rate Quote (pips)
Forward Rate
Forward Premium/Discount
Forward Contract Settlement Date
Forward Contract Size
Forward Contract Value
Forward Gain/Loss

175.0000
\$1.1675
+1.52%
December 29, 2024
€2,000,000
\$2,335,000
+\$35,000

185.0000
\$1.3185
+1.42%
December 29, 2024
£2,000,000
\$2,637,000
-\$37,000

Futures Contract Hedge

March 2020 Futures Contract
Futures Contract Position in FOREX
March 2020 Futures Contract Size
Number of March 2020 Futures Contracts
March 2020 Futures Hedge Position
March 2020 Futures Settlement Date
March 2020 Futures Settlement Price
Face Value of Futures Position
Net Futures Hedge Gain/Loss

Sell (EUR)
Short
€125,000
16
€2,000,000
December 18, 2024
\$1.1650
\$2,330,000
+\$30,000

Buy (GBP)
Long
£62,500
32
£2,000,000
December 18, 2024
\$1.3150
\$2,630,000
-\$30,000

Option Contract Hedge

March 2020 Option Contract
Option Contract Position
March 2020 Option Contract Size
Number of March 2020 Option Contracts
March 2020 Option Hedge Position
March 2020 Option Expiration Date
March 2020 Option Strike Price
March 2020 Option Premium
March 2020 Option Break-Even (Spot)
Cost of Option Hedge Position (premium)
Net Option Hedge "Floor" (Worst-Case)*
Net Option Hedge "Ceiling" (Best-Case)*

Put (EUR)
Short
€125,000
16
€2,000,000
December 13, 2024
\$1.1650
\$0.0200
\$1.1450 ↓
-\$40,000
-\$10,000
Unlimited

Call (EUR)
Long
£62,500
32
£2,000,000
December 13, 2024
\$1.3150
\$0.0050
\$1.3200 ↑
-\$10,000
-\$40,000
Unlimited (or EUR = 0)

* This "Net Option Hedge" is known only when the Option Contract either expires or is exercised. The "Floor" (worst-case) represents the highest possible cost of the net hedge position if the foreign currency moves against the original position and in favor of the option (which must be exercised to insure against loss in the original position). The "Ceiling" (best-case) represents the highest possible gain because the foreign currency moves in favor of the original position and the option simply expires unexercised. This is mostly theoretical because the foreign currency in a long position could increase in value to infinity but in a short position could only decrease in value to \$0.

Questions to consider:

1. What is the best hedge?

Note 1: In this simplified example, all the various hedge rates are the same (i.e., the Forward Rate, the Futures Settlement Rate, and the Option Strike Price). This equalizes the hedging alternatives in a way that may not exist in reality. Even though the Forward Rate and Futures Settlement Price should be roughly equal, they may not be identical as in this example.

Note 2: In this simplified example, we do not include the fees associated with each hedge. As has been discussed in class, each of the three hedging strategies would incur a “fee” that is paid to third parties (e.g., the Bank that agrees to the Forward Contract, the Broker that helps execute the Futures Contract, and the Broker that helps execute the Option Contract). In reality, these fees would be considered when deciding on a hedging strategy. Here we assume that the fees are roughly equivalent for each alternative and thus not included for consideration.

2. What problems are created by the transaction settlement date of the A/R and the settlement date of the Futures Contract Hedge? Same with the expiration date of the Options Hedge? Does this problem exist with a Forward Hedge?
3. What problems are created by the transaction settlement date of the A/P and the settlement date of the Futures Contract Hedge? Same with the expiration date of the Options Hedge? Does this problem exist with a Forward Hedge? Would it better to over or under time the A/P hedge?
4. What problems would be created by an A/R in the amount of €1,062,500 and the Futures Hedge? Same with the Options Hedge? Does this problem exist with a Forward Hedge? Would it be better to over or under hedge?
5. What problems would be created by an A/P in the amount of £1,031,250 and the Futures Hedge? Same with the Options Hedge? Does this problem exist with a Forward Hedge? Would it be better to over or under hedge?
6. What are the pros and cons of the Option hedge strategy? In particular, what forecast would be necessary to justify the Option strategy with the A/R? What forecast would be necessary to justify the Option strategy with the A/P?
7. Would it be wise to consider the A/R and the A/P to be roughly offsetting because the two underlying currencies are so highly correlated? How would that hedging strategy look? What would be the pros and cons of that strategy? What would be the greatest risk of that strategy?

Topic #6: Government Intervention

Foreign Exchange Intervention Definition, Strategies, Goals

By James Chen, CMT, an expert trader, investment adviser, and global market strategist
<https://www.investopedia.com/terms/f/foreign-exchange-intervention.asp>

What Is Foreign Exchange Intervention?

A foreign exchange intervention is a [monetary policy](#) tool that involves a central bank taking an active, participatory role in influencing the monetary funds transfer rate of the national currency, usually with its own reserves or its own authority to generate the currency. Central banks, especially those in developing countries, intervene in the [foreign exchange market](#) in order to build reserves for themselves or provide them to the country's banks. Their aim is often to stabilize the exchange rate.

- Foreign exchange intervention refers to efforts by central banks to stabilize a currency.
- Destabilizing effects can come from both market or non-market forces.
- Currency stabilization may require short-term or long-term interventions.
- Stabilization allows investors to be more comfortable with transactions using the currency in question.

Understanding Foreign Exchange Intervention

When a [central bank](#) increases the money supply through its various means of doing so, it must be careful to minimize unintended effects such as runaway inflation. The success of foreign exchange intervention depends on how the central bank [sterilizes](#) the impact of its interventions, as well as general macroeconomic policies set by the government.

Two difficulties that central banks face are determining the timing and amount of intervention, as this is often a judgment call rather than a cold, hard fact. The amount of reserves, the type of economic trouble facing the country, and the ever-changing market conditions require that a fair amount of research and understanding be in place before determining how to take a productive course of action. In some cases, a corrective intervention may have to be taken shortly after the first attempt.

Why Intervene?

[Foreign exchange intervention](#) comes in two flavors. Firstly, a central bank or government may assess that its [currency](#) has slowly become out of sync with the country's economy and is having adverse effects on it. For example, countries that are heavily reliant on [exports](#) may find that their currency is too strong for other countries to afford the goods they produce. They may intervene to keep the currency in line with the currencies of the countries which import their goods.

The [Swiss National Bank](#) (SNB) took this kind of action from September 2011 to January 2015. The SNB set a minimum exchange rate between the Swiss franc and the euro. This kept the Swiss franc from strengthening beyond an acceptable level for other European [importers](#) of Swiss goods. This approach was successful for three and half years after which the SNB determined that it had to let the Swiss franc float freely. Suddenly, without prior warning, the Switzerland central bank released the minimum exchange rate. This had highly negative consequences to some businesses, but, generally, the Swiss economy has been unfazed by the intervention.

Intervention can also be a short-term reaction to a certain event. A one-off event may cause a country's currency to move in one direction in a very short space of time. Central banks will intervene with the sole purpose of providing [liquidity](#) and reducing volatility. After the SNB lifted the floor in its currency against the Euro, the Swiss franc plummeted by as much as 25 percent. The SNB intervened in the short term to stop the Franc from falling further and curb the volatility.

Risks of Foreign Exchange Intervention

Foreign exchange interventions can be risky because they can undermine a central bank's credibility if it fails to maintain stability. Defending the national currency from speculation was a precipitating cause of the 1994 currency crisis in Mexico, and was a leading factor in the [Asian financial crisis](#) of 1997.



EMERGING MARKET PERSPECTIVES

The Case for Intervention

Under the right conditions, foreign exchange intervention can reduce unwarranted currency volatility

Sukudhew Singh

When central banks in the world's large economies slashed interest rates after 2008, smaller emerging market economies, especially in Asia, faced a flood of capital that caused their currencies to appreciate and interest rates to fall. Now that major central banks are rapidly tightening policy, financial flows have reversed: emerging market currencies are depreciating, inflation is increasing, and central banks are under pressure to raise interest rates even as growth stalls.

Global economic and financial integration has weakened the national transmission of monetary policy and made international factors a stronger driver of domestic prices and economic conditions. Free-floating currencies are ideal for most emerging market economies, but external developments can soon throw exchange rates out of kilter with economic fundamentals. Policy autonomy is guaranteed only if economies are strong enough to withstand volatile exchange rates and significant misalignment.

Intervention in the foreign exchange market allows policymakers to moderate the pace and

extent of currency appreciation or depreciation. It can also counter pressure on the exchange rate by reducing one-sided expectations about the currency's future value. A deeper financial system helps with intermediation but can be a double-edged sword: increased availability of financial instruments and greater liquidity may attract more capital inflows. Open emerging market economies with large globally integrated financial systems must hold more foreign exchange reserves and intervene more aggressively to avoid excessive volatility. Successful intervention is not guaranteed, however.

Successful intervention

Several factors make intervention more likely to work. For brevity, I will focus here on those that determine success when defending a depreciating currency.

- **Level of foreign exchange reserves:** Foreign reserves are not costless, but they are invaluable when the exchange rate comes under unwarranted depreciation pressure. They are even more important for countries with linked exchange rates (such as Hong Kong Special Administrative Region) or exchange-rate-based monetary frameworks (Singapore).
- **Strength of domestic economy and financial system:** The strength of these fundamentals provides the central bank with greater flexibility in terms of how much to intervene and let the exchange rate move. It allows more effective intervention because the central bank does not have to engage actively in liquidity operations that undermine its foreign exchange interventions (see the fourth bullet).
- **Intended exchange rate that is “defensible”** in that it reflects economic fundamentals: Pressure on the currency due to sustained outflows on the trade and current accounts of the balance of payments often reflects a failure to create a diversified, competitive, and globally integrated economy. Intervention will not help. If weak domestic fundamentals such as a large fiscal deficit, excessive monetary growth, or high inflation are affecting the exchange rate, then intervention would also be futile. Unless there is a determined effort to deal with these weaknesses, they will continue to exert a negative influence on the currency.
- **Actions by central banks to manage the liquidity consequences of intervention:** When the central bank intervenes to defend the exchange rate, it decreases the supply of the local currency

and increases the supply of foreign currencies. If other factors remain the same, this should support the local currency's exchange rate. The decrease in local currency liquidity pushes up domestic interest rates, which provides additional support to the exchange rate. However, the central bank often wants to protect the domestic economy from higher interest rates. The government is also likely to be unhappy with the higher cost of financing public debt. So the central bank usually injects liquidity back into the banking system, keeping local interest rates relatively stable but undermining its efforts to support the currency. If a weaker currency is causing higher domestic inflation, these liquidity operations weaken not only the exchange rate but domestic price stability as well. This makes monetary policy and the intervention operations less effective.

- **Openness of the capital account:** Open emerging market economies vary in their degree of openness, particularly when it comes to the capital account. An open capital account can facilitate two-way flows under normal circumstances, but large one-way flows during times of instability can overwhelm the central bank's ability to stabilize the currency. Yet it is crucial to avoid large swings in the exchange rate because of the ease with which short-term financial flows by residents and nonresidents can occur in response to exchange rate expectations.
- **Private sector foreign currency exposure and how much it is hedged:** In emerging market economies, the central bank must track this exposure carefully and even regulate it to ensure that it poses no risk to national economic and financial stability. Without these precautions, pressure on the exchange rate from panicked buying of foreign exchange can negate interventions to support the currency.

Adequacy of reserves

The level of reserves is important not only for intervention but also for instilling confidence in a country's ability to pay its way in the world. Maintaining a sufficiently large stock of reserves is an important policy consideration.

One way to reduce demand on the central bank's reserves is to **develop the local foreign currency market**, providing more opportunity for private intermediation of foreign exchange flows and for new hedging instruments. This should reduce the

frequency of central bank intervention. A common problem in times of uncertainty is that foreign currency dries up due to excess demand or hoarding. Ultimately, the central bank's reserves must again provide the safety mechanism for the market.

The sustainability of reserves also **depends on the sources** from which they are built. Reserves built from current account surpluses and flows of foreign direct investment are generally more reliable than reserves from short-term portfolio flows. Reserves should be built during good times. Central banks in emerging market economies are often vulnerable to political pressure that diverts existing reserves to other purposes. This leaves countries vulnerable and limits central banks' capacity to intervene when they need to.

There are **emergency sources of reserves**. Funding from the IMF is an option, but it is an option of last resort for many countries, especially in Asia. Countries also have bilateral swap arrangements to provide emergency liquidity in dollars or local currencies. Among the ASEAN+3 economies, a \$240 billion resource-pooling arrangement known as the Chiang Mai Initiative Multilateralisation Agreement provides liquidity support to regional economies in times of external stress. However, it has not diminished member economies' desire to build their own reserves for various reasons, including policy independence.

When reserves are running low, or capital flows are so large that intervention is unlikely to succeed, more direct intervention is needed to restore stability. Policymakers can rightfully consider **measures to restrict financial flows**. Many factors that make for successful intervention will also come into play in determining the success of capital controls. Policymakers who impose capital controls must also be cautious in timing their removal—doing so prematurely can be as risky as keeping them in place too long.

Done right, capital controls can act as a circuit breaker to preserve foreign reserves and provide policymakers with temporary breathing room for reforms to reduce vulnerabilities and support the economy, without the worry of external instability. Confidence in the local economy is a key fundamental that must be restored through credible policies, after which the controls can be gradually relaxed and removed. **FD**

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Topic 7: International Arbitrage

Arbitrage can be loosely defined as capitalizing on a discrepancy in quoted prices by making a riskless profit. In many cases, the strategy involves no risk (because it is simultaneous) and does not require that funds be tied up. The type of arbitrage discussed here applies to foreign exchange and international money markets and takes three common forms:

1. locational arbitrage
2. triangular arbitrage
3. covered interest arbitrage

Locational Arbitrage

Commercial banks providing foreign exchange services normally quote roughly the same rates on currencies, so shopping around may not lead to a more favorable rate. If the demand and supply conditions for a particular currency vary among banks then a given currency may be priced at different rates, in which case market forces will lead to realignment.

When quoted exchange rates vary among locations, participants in the foreign exchange market can capitalize on the discrepancy. Specifically, they can use locational arbitrage, which is the process of buying a currency at a location where it is priced cheap and then immediately selling it at some other location where it is priced higher.

ASSESSMENT

Assess the Opportunity

Step 1: Compare the Ask Rate in a foreign currency at one bank (Bank A) with the Bid Rate in the same currency at another bank (Bank B).

If you can purchase the currency from Bank A (at the Ask Price) and sell it to Bank B (at the Bid Price) for a profit (called a “round-trip transaction”), you have a locational arbitrage opportunity.

CALCULATING PROFITS/LOSSES

Step 1: Divide the amount of USD (that can be deployed in the arbitrage) by the Ask Price (quoted by the bank that sells you the currency) to calculate the total foreign currency purchased.

Step 2: Multiply the amount of foreign currency purchased in Step 1 by the Bid Price (quoted by the bank that buys the currency from you) to calculate the amount of USD that returns to you in the round-trip transaction.

Step 3: Subtract the amount in Step 2 from the amount in Step 1 to determine the profit.

Example:	The amount of USD that can be deployed for arbitrage is \$1,000,000
Step 1:	The Ask Price of EUR quoted by Bank A is \$1.2535, thus €797,466 is purchased
Step 2:	The Bid Price of EUR quoted by Bank B is \$1.2616, thus \$1,006,462 is returned
Step 3:	The profit is \$6,462 (\$1,006,462 - 1,000,000)

Note: This example does not include any assumptions for transactions fees and taxes

Triangular Arbitrage

A “cross-exchange rate” is the relationship between two currencies that is determined by their exchange rates with a common third currency. In the United States, the term cross-exchange rate refers to the relationship between two non-USD currencies based on the exchange rate of each currency with the USD. If a spot rate between two currencies differs from their cross-exchange rate using USD, an arbitrager can capitalize on the discrepancy because the cross-rate determines that one currency is undervalued (in terms of USD) and another currency is overvalued (in terms of USD) when compared to the spot rate. The undervalued currency can be used to acquire the overvalued currency, which can then be used to acquire USD, resulting in a profit to the arbitrager.

ASSESSMENT

Assess the Opportunity

Step 1: Calculate the cross-rate of any two currencies by using quoted spot rates (in USD) of the foreign currencies. **Note: See the Handout: “Simple Rules for Currency Cross Exchange Rates” for instructions on how to calculate cross-rates by using spot rates.**

Step 2: Compare the cross-rates calculated in Step 1 with the spot rate of those two currencies as quoted by a bank (the spot rate of the foreign currencies can be expressed in either currency).

Step 3: If a material discrepancy exists between the cross-rate calculated in Step 1 and the quoted spot rate retrieved in Step 2, you have found a triangular arbitrage opportunity. The discrepancy will cause one of the currencies to be “overvalued” (in terms of USD) and the other to be “undervalued” (in terms of USD) based on the spot rate of the two currencies with each other. This disequilibrium is what creates the opportunity because the **overvalued** currency can be purchased with USD, then exchanged for the **undervalued** currency, and finally (to complete the “round trip”) the **undervalued** currency can be used to purchase USD.

CALCULATING PROFITS/LOSSES

Step 1: Once the overvalued currency has been identified in the Assessment, divide the amount of USD that can be deployed to purchase the **overvalued** currency by the spot rate (direct quote).

Step 2: Use the **overvalued** currency (purchased in Step 1) to purchase the **undervalued** currency at the spot rate. (Note: This calculation depends on which currency was purchased and which currency is the base currency in the spot rate.)

Step 3: Multiply the currency purchased in Step 2 by the spot rate (direct quote) of that currency with the USD.

Step 4: Subtract the amount of USD acquired in Step 3 with the amount of USD deployed in Step 1.

Example

<u>Spot Quotes</u>	<u>Spot</u>	<u>Portfolio =</u>	<u>\$1,000,000</u>	
USD/GBP	\$1.5447	→	£647,375	USD into GBP
EUR/GBP	€ 1.3713	→	€ 887,745	GBP into EUR
USD/EUR	\$1.1356	→	\$1,008,123	EUR into USD
			\$8,123	Profit 0.81%
<u>Crossrate Calculation</u>				
EUR/GBP	€ 1.3603	→	€ 880,592	
USD/EUR	\$1.1356	→	\$1,000,000	
			No Profit	

Note: This example does not include any BID/ASK spreads of the currencies, and it does not include any assumptions for transactions fees and taxes.

Covered Interest Arbitrage

ASSESSMENT

Assess the Opportunity with regard to the Interest Rate Spread (conventional carry-trade)

Step 1: Compare the Loan Rate (the rate at which money can be borrowed) in one currency with the CD Rate (the rate at which money can be invested) in another currency to determine where the Interest Rate Spread may present an opportunity for profits.

Step 2: Calculate Spot/Forward Spread in the same two currencies.

Step 3: Compare the Interest Rate Spread with the Spot/Forward Spread to determine whether or not the Interest Rate gain is offset by a Spot/Forward loss. If the Interest Rate Spread exceeds the offsetting Spot/Forward Spread, an arbitrage opportunity exists. *Note: In normal conditions, the bid/ask spread, transaction fees, and taxes would be relevant components of the calculation.*

Assess the Opportunity with regard to the Spot/Forward Spread (unconventional)

Step 1: Calculate the Spot/Forward Spread for two currencies to determine which currency is appreciating and may present an opportunity for profits

Step 2: Compare the Loan Rate in the appreciating currency with the CD Rate in the depreciating currency to determine the Interest Rate Spread

Step 3: Compare the Interest Rate Spread with the Spot/Forward Spread to determine whether or not the forward gain in the appreciating currency is offset by the loss in the Interest Rate Spread. If the Spot/Forward gain in the appreciating currency is not completely offset by the Interest Rate loss, an arbitrage opportunity exists. *Note: In normal conditions, the bid/ask spread, transaction fees, and taxes would be relevant components of the calculation.*

CALCULATING PROFITS/LOSSES

Step 1: Calculate the amount borrowed

Step 2: Calculate the amount DUE (principal + interest) upon maturity of the loan (use Loan Rate)

Step 3: Convert the amount borrowed (Step 1) into the investment currency (use the Spot Rate) to determine the amount invested.

Step 4: Calculate the amount RECEIVED upon maturity of the investment (use CD Rate)

Step 5: Convert the investment RECEIVED into the borrowed currency (use Forward Rate)

Step 6: Compare the amount RECEIVED (Step 5) with the amount DUE (Step 2)

Step 7: Determine profit or loss (in the borrowed currency)

If necessary, convert profit or loss back into the home currency

If desired, calculate % gain/loss (divide profit or loss by the original amount borrowed)

Covered Interest Arbitrage

		Bank Interest Rates					
		<u>CD</u>	<u>Loan</u>				
	Zurich	4.50%	5.00%				
	London	4.25%	4.75%				
	Frankfurt	4.00%	4.50%				
	New York	1.50%	2.00%				
		<u>Borrowing Capacity</u>					
	Zurich	SFr. 10,000,000					
	London	£7,500,000					
	Frankfurt	€ 10,000,000					
	New York	\$10,000,000					
		FX Market Data			Borrow in USD : Invest in FX		
			12-month	Spot/Forward	Interest Rate	USD/Int. Rate	
	<u>Currency</u>	<u>Spot</u>	<u>Forward</u>	<u>Spread</u>	<u>Spread</u>	<u>Differential</u>	
	Swiss Franc	\$0.9562	\$0.9227	-3.50%	2.50%	-1.00%	
	British Pound	\$1.5447	\$1.4945	-3.25%	2.25%	-1.00%	
	Euro	\$1.1356	\$1.1242	-1.00%	2.00%	1.00%	
		FX Market Data			Borrow in FX : Invest in USD		
			12-month	Spot/Forward	Interest Rate	FX/Int. Rate	
	<u>Currency</u>	<u>Spot</u>	<u>Forward</u>	<u>Spread</u>	<u>Spread</u>	<u>Differential</u>	
	Swiss Franc	\$0.9562	\$0.9227	-3.50%	-3.50%	0.00%	
	British Pound	\$1.5447	\$1.4945	-3.25%	-3.25%	0.00%	
	Euro	\$1.1356	\$1.1242	-1.00%	-3.00%	-2.00%	
	<u>Loan Portfolio</u>	<u>NY to Zurich</u>	<u>NY to London</u>	<u>NY to Frankfurt</u>	<u>Zurich to NY</u>	<u>London to NY</u>	<u>Frankfurt to NY</u>
Step 1	Borrow	\$10,000,000	\$10,000,000	\$10,000,000	SFr. 10,000,000	£7,500,000	€ 10,000,000
	<u>Interest charged</u>	<u>\$200,000</u>	<u>\$200,000</u>	<u>\$200,000</u>	<u>SFr. 500,000</u>	<u>£356,250</u>	<u>€ 450,000</u>
Step 2	Due (P & I)	\$10,200,000	\$10,200,000	\$10,200,000	SFr. 10,500,000	£7,856,250	€ 10,450,000
	<u>Investment Portfolio</u>						
Step 3	Convert at spot	SFr. 10,458,063	£6,473,749	€ 8,805,918	\$9,562,000	\$11,585,250	\$11,356,000
	<u>Interest gained</u>	<u>SFr. 470,613</u>	<u>£275,134</u>	<u>€ 352,237</u>	<u>\$143,430</u>	<u>\$173,779</u>	<u>\$170,340</u>
Step 4	Total (P & I)	SFr. 10,928,676	£6,748,883	€ 9,158,154	\$9,705,430	\$11,759,029	\$11,526,340
Step 5	Convert at forward	\$10,084,250	\$10,086,188	\$10,296,000	SFr. 10,518,135	£7,868,217	€ 10,252,525
Step 6	<u>Repay Loan</u>	<u>\$10,200,000</u>	<u>\$10,200,000</u>	<u>\$10,200,000</u>	<u>SFr. 10,500,000</u>	<u>£7,856,250</u>	<u>€ 10,450,000</u>
Step 7	Difference	-\$115,750	-\$113,813	\$96,000	SFr. 18,135	£11,967	-€ 197,475
	Profit/Loss	-\$115,750	-\$113,813	\$96,000	SFr. 18,135	£11,967	-€ 197,475

Note: This example does not include any BID/ASK spreads of the currencies, and it does not include any assumptions for transactions fees and taxes.

Simple Rules for Currency Cross Exchange Rates

A crossrate calculation is the process of using the exchange rates of two currencies (i.e., € and £) with a common third currency (i.e., \$) to determine the exchange rate between the two currencies.

Example 1:

- | | |
|---|---|
| A. Direct quote of the USD exchange rate with the British Pound | $\$1.6364 = \$ \text{ to } \pounds 1 \text{ or } \$/\pounds$ |
| B. Direct quote of the USD exchange rate with the Euro | $\$1.2849 = \$ \text{ to } \text{€} 1 \text{ or } \$/\text{€}$ |
| C. The crossrate of Euros per British Pound (A divided by B) | $\text{€} 1.2736 = \text{€} \text{ to } \pounds 1 \text{ or } \text{€}/\pounds$ |
| D. The crossrate of British Pounds per Euro (B divided by A) | $\pounds 0.7852 = \pounds \text{ to } \text{€} 1 \text{ or } \pounds/\text{€}$ |

Example 2:

- | | |
|---|---|
| A. Direct quote of the USD exchange rate with the British Pound | $\$1.6364 = \$ \text{ to } \pounds 1 \text{ or } \$/\pounds$ |
| B. Indirect quote of the USD exchange rate with the Euro | $\text{€} 0.7783 = \text{€} \text{ to } \$ 1 \text{ or } \text{€}/\$$ |

The crossrate of Euros per British Pound (A times B)	$\text{€} 1.2736 = \text{€} \text{ to } \pounds 1 \text{ or } \text{€}/\pounds$
--	---

Example 3:

- | | |
|---|--|
| A. Indirect quote of the USD exchange rate with the British Pound | $\pounds 0.6111 = \pounds \text{ to } \$ 1 \text{ or } \pounds/\$$ |
| B. Direct quote of the USD exchange rate with the Euro | $\$1.2849 = \$ \text{ to } \text{€} 1 \text{ or } \$/\text{€}$ |

The crossrate of Euros per British Pound (A times B)	$\pounds 0.7852 = \pounds \text{ to } 1\text{€} \text{ or } \pounds/\text{€}$
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Rules for calculating crossrates:

- If the two rates (against a common currency) are in the same form (Example 1): **DIVIDE**
(i.e., both are direct quotes in the third currency: \$/£ and \$/€)
- If the two rates (against a common currency) are in different forms (Examples 2 and 3): **MULTIPLY**
(i.e., one is a direct quote in the third currency \$/£, the other is an indirect quote)

Other helpful hints:

- When calculating a crossrate using two direct quotes through division, the currency in the denominator of the division will be the resulting direct quote against the currency in the numerator.

Example 4:

You have two direct USD quotes (\$/£ and \$/€) and you divide the USD/GBP quote by the USD/EUR quote

$$(\$/\pounds \div \$/\text{€}) = (\$/\pounds \times \text{€}/\$) = \text{€}/\pounds \quad \text{or} \quad \$1.6364 \div \$1.2849 = \text{€}1.2736$$

Example 5:

You have two direct USD quotes (\$/£ and \$/€) and you divide the USD/EUR quote by the USD/GBP quote

$$(\text{€}/\$ \div \$/\pounds) = (\text{€}/\$ \times \pounds/\$) = \pounds/\text{€} \quad \text{or} \quad \$1.2849 \div \$1.6364 = \pounds 0.7852$$

Triangular Arbitrage In-Class Problems

	<u>Spot Quotes</u>
USD per GBP	\$1.2214
USD per EUR	\$1.1018
EUR per GBP	€1.2085

1. Does the cross-rate calculation equal the quoted Spot Rate?
2. If not, which currency (EUR or GBP) is overvalued in the Spot Rate?
3. Describe the round trip arbitrage:
4. Calculate the round trip arbitrage using \$1,000,000

	<u>Spot Quotes</u>
USD per GBP	\$1.2214
USD per EUR	\$1.1018
EUR per GBP	€1.0085

1. Does the cross-rate calculation equal the quoted Spot Rate?
2. If not, which currency (EUR or GBP) is overvalued in the Spot Rate?
3. Describe the round trip arbitrage:
4. Calculate the round trip arbitrage using \$1,000,000

Triangular Arbitrage

Answers to In-Class Problems

	<u>Spot Quote</u>
USD per GBP	\$1.2214
USD per EUR	\$1.1018
EUR per GBP	€1.2085

1. Does the cross-rate calculation equal the quoted Spot Rate?

The cross-rate calculation that sets EUR per GBP would be $\$1.2214 \div \$1.1018 = \text{€}1.1085$, which does not equal the quoted rate of €1.2085

2. If not, which currency (EUR or GBP) is overvalued in the Spot Rate?

The GBP is overvalued because £1 is purchasing €1.2085 in the Spot Rate but should only purchase €1.1085 based on the cross-rate calculation

3. Describe the round trip arbitrage:

- a. Use USD to buy the overvalued currency (GBP)
- b. Use the overvalued currency (GBP) to buy the undervalued currency (EUR)
- c. Use the undervalued currency (EUR) to buy USD

4. Calculate the round trip arbitrage using \$1,000,000

- a. $\$1,000,000 = \text{£}818,733 @ \1.2214
- b. $\text{£}818,733 = \text{€}989,438 @ \text{€}1.2085$
- c. $\text{€}989,438 = \$1,090,163 @ \1.1018
- d. Profit = \$90,163 on a \$1,000,000 investment (9.02%)

	<u>Spot Quote</u>
USD per GBP	\$1.2214
USD per EUR	\$1.1018
EUR per GBP	€1.0085

1. Does the cross-rate calculation equal the quoted Spot Rate?

The cross-rate calculation that sets EUR per GBP would be $\$1.2214 \div \$1.1018 = \text{€}1.1085$, which does not equal the quoted rate of €1.0085

2. If not, which currency (EUR or GBP) is overvalued in the Spot Rate?

The EUR is overvalued because £1 is purchasing €1.0085 in the Spot Rate but should purchase €1.1085 based on the cross-rate calculation

3. Describe the round trip arbitrage:

- a. Use USD to buy the overvalued currency (EUR)
- b. Use the overvalued currency (EUR) to buy the undervalued currency (GBP)
- c. Use the undervalued (GBP) currency to buy USD

4. Calculate the round trip arbitrage using \$1,000,000

- a. $\$1,000,000 = \text{€}907,606 @ \1.1018
- b. $\text{€}907,606 = \text{£}899,956 @ \text{€}1.0085$
- c. $\text{£}899,956 = \$1,099,206 @ \1.2214
- d. Profit = \$99,206 on an investment of \$1,000,000 (9.92%)

Topic 8: International Parity Relationships and Forecasting

PURCHASING POWER PARITY - PPP

An economic theory that attempts to quantify the relationship between inflation and exchange rates based on relative rates of inflation in two countries. PPP supports the notion that relatively high inflation places downward pressure on a currency value and estimates the amount of adjustment needed on the exchange rate between countries in order for the exchange to be equivalent to each currency's purchasing power. In other words, the exchange rate adjusts so that an identical good in two different countries has the same price when expressed in the same currency. For example, a chocolate bar that sells for C\$1.50 in a Canadian city should cost US\$1.00 in a U.S. city when the exchange rate between Canada and the U.S. is 1.50 CDN to USD. (Both cost US\$1.00.)

Absolute Form of PPP is based on the idea that, in the absence of international barriers, consumers will shift their demand to wherever prices are lowest, and if there is a discrepancy in the prices as measured by a common currency, demand will shift so that these prices converge. **Relative Form** of PPP accounts for market imperfections (transportation, tariffs, quotas) to make it unlikely for prices in different countries to be perfectly identical, but does suggest that the rate of change in prices should be comparable when measured in a common currency.

INTERNATIONAL FISHER EFFECT - IFE

An economic theory that states that an expected change in the current exchange rate between any two currencies is approximately equivalent to the difference between the two countries' nominal interest rates for that time. For example, if country A's interest rate is 10% and country B's interest rate is 5%, country B's currency should appreciate roughly 5% compared to country A's currency.

The rationale for the IFE is that a country with a higher interest rate will also tend to have a higher inflation rate (i.e., expected rates of inflation are the primary factors in setting interest rates). This increased amount of inflation should cause the currency in the country with the high interest rate to depreciate against a country with lower interest rates.

INTEREST RATE PARITY - IRP

A theory in which the interest rate differential between two countries is equal to the differential between the forward exchange rate and the spot exchange rate. Interest rate parity plays an essential role in foreign exchange markets by connecting interest rates, spot exchange rates, and forward exchange rates. IRP should negate the possibility of a Currency Carry Trade based on Covered Interest Arbitrage.

CURRENCY CARRY TRADE

A strategy in which an investor borrows and sells a certain currency with a relatively low interest rate and uses the funds to purchase and invest a different currency yielding a higher interest rate. A trader using this strategy attempts to capture the difference between the rates, which can often be substantial.

For example, if Japan's interest rates are low but Australia's interest rates are high, a speculator might borrow yen (at low cost), convert to Australian dollars, invest in Australia, and leave the position "unhedged" – i.e., no locked in forward rate. If the difference in the interest rates is not offset by the movement of the exchange rates, the speculator gains.

Examples of Purchasing Power Parity (PPP) and International Fisher Effect (IFE)

Purchasing Power Parity

Assume that the United States and the United Kingdom trade extensively with each other and initially have zero inflation. Now suppose that the United States experiences a 9% inflation rate while the United Kingdom experiences a 5% inflation rate. Under these conditions, PPP theory suggests that the GBP should appreciate by approximately 4% (the difference between their inflation rates). Given British inflation of 5% and the GBP's appreciation of 4%, U.S. consumers will have to pay about 9% more for British goods than they paid in the initial equilibrium state. That value is equal to the 9% increase in prices of U.S. goods due to U.S. inflation. The exchange rate should adjust to offset the differential in the two countries' inflation rates, in which case the prices of goods in the two countries should appear similar to consumers.

Reconsider the previous example but now suppose that the GBP appreciated by only 1% in response to the inflation differential. In this case, the increased price of British goods to U.S. consumers will be approximately 6% (5% inflation and 1% appreciation in the GBP), which is less than the 9% increase in the price of U.S. goods to U.S. consumers. We should therefore expect U.S. consumers to continue shifting their consumption to British goods. Purchasing Power Parity (PPP) suggests that this increased U.S. consumption of British goods by U.S. consumers would persist until the GBP appreciated by about 4%. Thus, from the U.S. consumer's viewpoint, any level of appreciation lower than this would result in lower British prices than U.S. prices.

From the British consumer's viewpoint, the price of U.S. goods would have initially increased by 4% more than British goods. Hence British consumers would continue to reduce their imports from the U.S. until the GBP appreciated enough to make U.S. goods no more expensive than British goods. The net effect of the GBP appreciating by 4% is that the prices of U.S. goods would increase by approximately 5% to British consumers (9% inflation minus the 4% savings to British consumers due to the GBP's 4% appreciation).

International Fisher Effect

Suppose that inhabitants of the United States are willing to save money only if they can earn a real interest rate of 2%. This means that their savings must grow by 2% more than the general price level of products that savers might purchase in the future with their savings. If the expected U.S. annual rate of inflation is 1%, then the one-year interest rate on a savings deposit should be 3% in order to allow for a real interest rate of 2%.

As expected inflation in the United States changes over time, the nominal interest rate would have to change as well in order to provide savers with a real interest rate of 2%. Suppose economic conditions have led U.S. savers to believe that the expected annual rate of inflation is now 4% instead of 1%. At the nominal interest rate of 3%, saving would be a losing proposition because the prices of products for which consumers are saving increase more rapidly than their savings. If the nominal interest rate on deposits remained lower than expected inflation, then the real interest rate would be negative; in this case, savers may be better off spending their money now (and even borrowing, if necessary) in order to purchase products before prices increase. Thus savers are willing to save money only if the nominal interest rate on savings deposits is 6%, which would allow for a 2% real rate of interest (after accounting for the expected annual inflation of 4%). This means that financial institutions must increase their deposit rate to 6% in order to attract savers.

Now assume that economic conditions change again, causing savers to expect that annual inflation will be 6% percent over the next year. If the prevailing nominal interest rate of 6% did not change, then savers would be unwilling to save money because prices are increasing as fast as the rates paid on savings deposit. Under these conditions, savers would be willing to save only if the nominal one-year interest rate on deposits rises to 8%; that would allow for a real rate of interest of 2% after accounting for the expected annual inflation of 6%.

Interest Rate Parity according to the Fisher Effect and the effects of PPP

Assume that the real rate of interest is 2% in both Canada and the United States, and assume that the one-year nominal interest rate is 13% in Canada versus 8% in the United States. According to the Fisher Effect, the expected inflation rate over the next year in each country is equal to the nominal interest rate minus the real rate of interest. For Canada, the expected inflation rate is $13\% - 2\% = 11\%$ and for the United States it is $8\% - 2\% = 6\%$. Hence the difference in expected inflation between the two countries is $11\% - 6\% = 5\%$. Note that this difference in expected inflation between the countries is equal to the difference in nominal interest rates ($13\% - 8\% = 5\%$) between them.

Recall from the preceding example that Canada has an expected inflation of 11% versus 6% in the U.S. Since the expected inflation is 5% higher in Canada, PPP suggests that the Canadian dollar should depreciate against the U.S. dollar by about 5%.

Observe that this level of depreciation would offset the interest rate advantage in Canada, so that U.S. investors would not benefit from investing there. They would earn about 8% on a savings deposit in Canada if the Canadian dollar weakened by 5% over the year, which is similar to what they would have earned in the United States.

Suppose the real rate of interest is 2% in each country. The U.S. inflation rate is expected to be 6% while the inflation rate in Japan is expected to be 3%.

According to PPP theory, the Japanese yen is expected to appreciate by the expected inflation differential of 3%. If the exchange rate changes as expected, then Japanese investors who attempt to capitalize on the higher U.S. interest rate will earn a return similar to what they could have earned in their own country. Although the U.S. interest rate is 3% higher than the Japanese interest rate, the Japanese investors will repurchase their yen at the end of the investment period for 3% more than the price at which they initially exchanged yen for dollars. Therefore, their return from investing in the United States is no better than what they would have earned domestically.

Suppose the nominal interest rate is 13% in Canada and 5% in Japan, and suppose the expected real rate of interest is 2% in each country. The expected inflation differential between Canada and Japan is 8%. According to PPP theory, this inflation differential means that the Canadian dollar should depreciate by 8% against the yen. Therefore, even though Japanese investors would earn an additional 8% interest on a Canadian investment, the Canadian dollar would be valued at 8% less by the end of the period. Under these conditions, the Japanese investors would earn a return of about 5% by investing in Canada, which is the same as what they would earn on an investment in Japan.

Topic 9: Managing Transaction Exposure: Money Market Hedge

Money Market Hedge

A money market hedge involves taking a **money market position** (short-term investing, short-term borrowing, or both) to cover foreign currency transaction risk (created by an A/R or A/P in the foreign currency). The company can create a simplified (called a “**single-position**”) money market hedge by establishing a money-market position in the foreign currency when the transaction is booked. This requires investing in the money market to hedge a Payable, and borrowing in the money market to hedge a Receivable.

Since companies prefer to hedge without using their cash balances, especially when the transaction being hedged is a Payable, a money market hedge can be used by establishing two money market positions (called a “**double-position**” hedge). A double-position A/P hedge means the company:

- (1) borrows funds in USD in the US money market; and
- (2) converts the borrowed funds into the foreign currency and invests them in the foreign money market so that the invested funds are eventually used to cover the Payable.

When the company’s transaction risk involves an A/R, it can create a single-position money market hedge that allows it to borrow against the Receivable. This means the company receive cash immediately, although at a discounted value compared to the Book Value of the A/R (which means this hedge results in a guaranteed “loss” on the A/R). Although it is not as common in the case of an A/R, the company can use a double-position hedge to either reduce or eliminate (depending on the interest rate spread) the “loss” from the single-position A/R hedge.

Money Market Hedge on a Payable (A/P): Single-position

A money market hedge on an A/P involves taking a money market position (i.e., short-term borrowing) to cover a future Payable. If a firm has excess cash, the simple money market hedge (called a “single-position”) is created by investing in a foreign bank an amount of funds that will be sufficient (with interest made on the investment) to cover the A/P on the settlement date.

EXAMPLE: If Ravencorp needs £2,000,000 in 180 days to cover an A/P, it could convert USD to GBP today and invest these EUR in a 180-day time deposit. Assuming that the APR in Great Britain on time deposits is 5.00%, the 180-day investment would earn 2.50% interest. Thus, Ravencorp would need to invest £1,951,220 in order to have £2,000,000 upon maturity. The calculation for this amount is (.025 is the periodic rate):

$$\text{Investment needed to hedge the A/P} = \frac{\text{£}2,000,000}{1 + .025} = \text{£}1,951,220$$

Assuming a spot rate of \$1.2800, the amount of USD needed to make the GBP investment today can be calculated as

$$\text{GBP investment in USD} = \text{£}1,951,220 \times \$1.28 = \$2,467,561$$

If the spot rate was \$1.2800 when the A/P was booked, the Book Value of the A/P is \$2,560,000, which is higher than the amount required for the GBP investment in the money market hedge. Thus, the money market hedge would result in a gain of \$62,439. However, to enjoy this gain, the company would have to use its cash today rather than wait 180 days (which is the term on the A/P).

Money Market Hedge on a Payable (A/P): Double-position

If Ravencorp did not want to tie up its own cash when investing in the GBP money market to hedge the Payable, it could borrow the amount of USD needed to make the GBP investment. Assume the APR in the US is 3.00%, which means the cost of borrowing for 180 days is 1.50%. The original USD loan required for the money market hedge is \$2,467,561, which means the amount needed to repay the USD loan (principal + interest) is calculated as follows (.015 represents the 1.50% interest on a 180-day loan):

$$\text{Dollar amount of loan repayment} = \$2,497,561 \times (1 + .015) = \$2,534,024$$

Thus, the money market hedge would result in a gain of \$24,976, which is less than the gain from the single-position hedge, but allowed the company to hold onto its cash for the 180 days.

Questions when calculating the Money Market Hedge of an A/P:

What is the spot rate?

What is the book value of the A/P?

What is the forward rate?

What is the gain/loss on the A/P using the forward rate? (This would be your forward hedge.)

What is the investing rate in the foreign currency?

What is the settlement value of the A/P using the single-position money market hedge?

What is the borrowing rate in the domestic currency?

What is the settlement value of the A/P using the double-position money market hedge?

What are the advantages and disadvantages of the money market hedge with only one position?

What are the advantages and disadvantages of the money market hedge with two positions?

Money Market Hedge: Single and Double Position (A/P)									
			<u>Hypothetical A/P</u>						
			A/P in GBP	£2,000,000					
			Spot Rate	\$1.2800					
			Book Value of A/P	\$2,560,000					
			Settlement Date	180 days					
			Interest Rate in UK	2.50%					Note: This is a periodic (180-day) rate. The APR is 5.00%
			Interest Rate in US	1.50%					Note: This is a periodic (180-day) rate. The APR is 3.00%
			Spread (UK over US)	1.00%					Forward rate should reflect GBP depreciating by this spread
			Forward Rate based on IRP	\$1.2672					
			Value of A/P using the forward rate	\$2,534,400					
			Gain/(Loss) on A/P using forward rate hedge	\$25,600				1.00% is the gain on the A/P	
			<u>Single-position hedge</u>						
			Amount due on A/P	£2,000,000					
			Amount invested in GBP	£1,951,220					This is the amount that must be invested in GBP so that the entire (Principal + Interest) of the investment matches the A/P when due
			Amount of accrued interest on GBP investment	£48,780					
			Principal + Interest on GBP investment	£2,000,000					This hedge requires the company to use this amount of cash to cover the A/P now, rather than wait 180 days to use its cash when A/P is due
			Amount of USD needed for GBP investment at Spot rate	\$2,497,561					
			Book Value of A/P	\$2,560,000					
			Gain/(Loss) on A/P	\$62,439				2.44% is the gain on the A/P	
			<u>Double-position hedge</u>						
			Amount invested in London (same as above)	£1,951,220					
			USD borrowed for GBP investment at Spot rate	\$2,497,561					By borrowing the amount that is needed for the GBP investment, the company does not use its own cash until the USD loan is due in 180 days. However, the interest on the USD loan lowers the gain on the A/P.
			Amount of accrued interest on borrowed USD funds	\$37,463					
			Amount due on borrowed US funds with interest	\$2,535,024					
			Book Value of A/P	\$2,560,000					
			Gain/(Loss) in double-position MM hedge	\$24,976				0.98% is the gain on the A/P	

Money Market Hedge on a Receivable (A/R): Single-position

A single-position money market hedge on a Receivable involves borrowing the foreign currency and then using the funds from the Receivables to pay off the loan. The due date on the foreign currency loan (principal + interest) should coincide with the settlement date of the Receivable.

EXAMPLE: Assume Ravencorp books an A/R of £2,000,000 due in six months (180 days), and the APR on borrowed funds in England is 5.00%. Thus, Ravencorp can borrow GBP from a London bank for 180 days at a periodic rate of 2.50%. The amount that Ravencorp borrows so that it can use all of its £2,000,000 Receivable to repay the entire loan in six months is calculated by this formula (.025 is the interest rate):

$$\text{Amount to borrow} = \frac{\text{£2,000,000}}{1 + .025} = \text{£1,951,220}$$

Note: If Ravencorp obtains a six-month loan of £1,951,220 and pays a 2.50% periodic rate, it will owe the bank £2,000,000 on the due date (including £48,780 of interest). This matches the A/R amount with the entire loan amount (principal + interest) that is due at the maturity.

The GBP funds that Ravencorp borrowed can be converted to USD and used to support existing operations. The amount of USD collected here depends on the spot rate at the time borrowed funds are received. Assume the spot rate is \$1.2800 when the A/R is booked (resulting in a Book Value of \$2,560,000). The amount borrowed in GBP (£1,951,220) is converted into \$2,497,561, which means the company loses \$62,439 on this transaction.

Money Market Hedge on a Receivable (A/R): Double-position

If the MNC does not need short-term funds to support existing operations, it can invest the funds received from the GBP loan in the US money market in order to reduce the loss taken on the A/R.

EXAMPLE: The GBP funds from the loan, converted to USD equals \$2,497,561. These funds are invested in the US money market in order to earn 1.50% over 180 days. The value of the USD investment at maturity is calculated as:

$$\$2,497,561 \times 1.015 = \$2,535,024$$

Thus, if Ravencorp uses the double-position money market hedge, its A/R will be worth \$2,535,024 in six months when the US investment matures with interest. This is still a loss of \$24,976 from the Book Value of the A/R (but less than the loss on the single-position money market hedge).

Questions when calculating the Money Market Hedge of an A/R:

What is the spot rate?

What is the book value of the A/R?

What is the forward rate?

What is the gain/loss on the A/R using the forward rate? (This would be your forward hedge.)

What is the borrowing rate in the foreign currency?

What is the value of the A/R using the single-position money market hedge?

What is the investing rate in the domestic currency?

What is the value of the A/R using the double-position money market hedge?

What are the advantages and disadvantages of the money market hedge with only one position?

What are the advantages and disadvantages of the money market hedge with two positions?

Money Market Hedge: Single and Double Position (A/R)				
Hypothetical A/R				
A/R in GBP	£2,000,000			
Spot Rate	\$1.2800			
Book Value of A/R	\$2,560,000			
Settlement Date	180 days			
Interest Rate in London	2.50%	Note: This is a periodic (180-day) rate. The APR is 5.00%		
Interest Rate in US	1.50%	Note: This is a periodic (180-day) rate. The APR is 3.00%		
Spread	1.00%	Forward rate should reflect GBP depreciating by this spread		
Forward Rate based on IRP	\$1.2672	GBP is depreciating in the 6-month forward		
Value of A/P using the forward rate	\$2,534,400			
Gain/(Loss) on A/P using forward rate hedge	\$25,600	1.01% is the gain on the A/R		
Single-position hedge				
Borrow in GBP Money Market (adj. for interest rate)	£1,951,220	This is the amount that can be borrowed so that the A/R is used to pay the entire amount (Principal + Interest) of the loan when it is due		
Interest Due on GBP Loan at Maturity (180 days)	£48,780			
Principal + Interest	£2,000,000			
Convert GBP borrowed into USD at Spot rate	\$2,497,561			
Book Value of A/R	\$2,560,000			
Gain/(Loss) on A/R	(\$62,439)	-2.44% is the loss on the A/R		
Double-position hedge				
Amount borrowed in London (same as above)	£1,951,220			
Convert funds borrowed at Spot rate and invest in US	\$2,497,561	This investment reduces the loss on the A/R, but it also ties up the cash until the Money Market investment matures.		
Accrued interest (180 days) on invested funds in US	\$37,463			
USD investment upon maturity w/interest	\$2,535,024			
Book Value of A/R	\$2,560,000			
Gain/(Loss) on A/R	(\$24,976)	-0.98% is the loss on the A/R		

Note on Interest Rate Spreads and Hedging Strategies:

Rule: If $I\%$ in GBP $>$ $I\%$ in USD, then IRP forces the GBP to depreciate (and USD to appreciate).

Forward hedge locks in a **gain** on an A/P in GBP

Forward hedge locks in a **loss** on an A/R in GBP

Rule: If $I\%$ in GBP $<$ $I\%$ in USD, then IRP forces the USD to depreciate (and GBP to appreciate).

Forward hedge locks in a **gain** on an A/R in GBP

Forward hedge locks in a **loss** on an A/P in GBP

Rule: The single-position Money Market hedge will always result in a **gain** on the A/P in any foreign currency

Note: The gain on the single-position A/P hedge will always be reduced or reversed with a double-position A/P hedge depending on the interest rate spread. This is due to the fact that the double-position requires borrowing the funds (in USD) that are converted and used to invest in the foreign currency.

Rule: The single-position Money Market hedge will always result in a **loss** on the A/R in any foreign currency

Note: The loss on the A/R can be reduced or even reversed with a double-position money market hedge, depending on the interest rate spread.

Money Market Hedge versus Forward Hedge Should a company use a forward contract hedge or a money market hedge? Since the results of both hedges are known beforehand, the firm can implement the one that is more feasible. If interest rate parity (IRP) holds and there are no transaction costs, then the money market hedge will yield the same results as the forward hedge. This is because the forward premium on the forward rate reflects the interest rate differential between the two currencies.

Topic 10: Translation Exposure

Companies that operate in multiple countries face translation risk when they consolidate the financial statements of all subsidiaries into one. A subsidiary's financial statement is normally measured in its local currency but must be "translated" into the currency of the parent company. Since exchange rates vary over time, the translation of the subsidiary's financial statement into a different currency is affected by exchange rate movements. The exposure of the company's consolidated financial statements to exchange rate fluctuations is known as translation exposure. Specifically, subsidiary earnings translated into the reporting currency on the consolidated Income Statement are subject to fluctuations in the exchange rates, as are net asset values when translated into the reporting currency on the consolidated Balance Sheet.

The extent of a company's translation exposure depends mainly on three factors:

1. the proportion of its business conducted by foreign subsidiaries,
2. the locations of its foreign subsidiaries, and
3. the accounting methods that it uses.

Proportion of Business by Foreign Subsidiaries: The greater the percentage of a company's business conducted by its foreign subsidiaries, the larger the percentage of a given financial statement item that is susceptible to translation exposure.

Locations of Foreign Subsidiaries: The countries in which subsidiaries are located can also influence the degree of translation exposure because the financial statement items of each subsidiary are typically measured by the respective subsidiary's home currency.

Accounting Methods: A company's degree of translation exposure is strongly affected by the accounting procedures used to translate when consolidating financial statement data. To translate earnings, MNCs use a process established by the Financial Accounting Standards Board (FASB). The guidelines for translation are set by FASB 52, which includes the following provisions:

1. The functional currency of an entity is the currency of the economic environment in which the entity operates.
2. The current exchange rate as of the reporting date is used to translate the assets and liabilities of a foreign entity from its functional currency into the reporting currency.
3. The weighted average exchange rate over the relevant period is used to translate revenues, expenses, and gains and losses of a foreign entity from its functional currency into the reporting currency.
4. Translated income gains or losses due to changes in foreign currency values are not recognized in current net income but are reported as a second component of stockholder's equity; an exception to this rule is a foreign entity located in a country with high inflation.
5. Realized income gains or losses due to foreign currency transactions are recorded in current net income, although there are some exceptions.

As noted above, under FASB 52, consolidated earnings are sensitive to the functional currency's weighted average exchange rate and consolidated asset values are affected by the exchange rate on the date of reporting.

Benedictine College School of Business

FINC-4900 International Finance

Fall 2026

MWF 9:00 – 9:50 am in Library 018

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MW: 2.00 pm – 3.00 pm
TR and all other times by appointment (use Navigate):
<https://benedictine.campus.eab.com/pal/mU3kdrHPhI>

Textbook: *International Finance Student Book of Handouts* (the \$15 course fee covers this item)

E-Textbook: *International Financial Management*, 13th Edition, Jeff Madura, Cengage Learning

Instructions: Go to the website: <http://www.cengagebrain.com/>
Click the “Students” box
Type into the search line the ISBN # of the book: **9780357689561**
You can purchase the eBook access until January 28 for \$76.99

Course Description: This course focuses on building a practical and theoretical understanding of international financial management. Topics include exchange rate determination, governmental influence on exchange rates, international arbitrage, direct foreign investment and multi-national cost of capital and capital budgeting. Prerequisite: FINC-3100 (Principles of Finance)

General Course Objectives:

- Develop an understanding of the International Monetary System, Foreign Exchange Markets, World Financial Markets, and the foreign monetary exchange issues involved in managing a Multinational Firm.
- Develop competence in Foreign Exchange (FOREX), Foreign Currency Risk Management, International Capital Budgeting, and International Risk Management.

Specific Course Objectives: Students should learn how to:

1. Identify and understand the unique issues associated with conducting international business.
2. Identify and understand the unique risks associated with international finance.
3. Read and use foreign currency exchange rates, both spot rates and forward rates.
4. Understand and use foreign currency forward contracts to mitigate risk
5. Understand and use foreign currency futures and option contracts to mitigate risk
6. Understand how governments can and do influence currency exchange rates.
7. Understand and calculate international arbitrage using exchange rates and interest rates
8. Understand and calculate parity relationships relative to foreign currency exchange rates
9. Identify, understand, and calculate international translation risk
10. Identify, model, and analyze the unique features of international capital budgeting

Teaching Method: To achieve these objectives, this course will use a combination of lecture, class discussion, and problem-solving covering the assigned reading material. In addition, students will complete homework assignments and projects that utilize their increasing knowledge of international finance as the course proceeds. To succeed in this class, the student should:

- Read the assigned material prior to coming to class
- Attend all classes, pay attention, and participate in class discussion
- Complete the ten homework assignments and two projects on time and produce good work
- Prepare and perform well on all graded material

Class Attendance: *It is presumed that students attend every session of courses in which they are registered in order that mastery of subject matter may be assisted by the instructor's leadership, class discussion, and/or class projects, and so that they may contribute to the intellectual exchange that is appropriate to the class. The College's mission of education within a community of faith and scholarship makes class attendance a particular priority. However, it is recognized that in some cases absence is unavoidable. Sickness excuses are received by the individual instructors. The Director of the Student Success Center will forward information to faculty about long-term absences or absences that may be considered unavoidable. Individual instructors have the right to limit absence in their courses and impose penalties for absenteeism, providing the professor's policy and attached penalties are identified in the course syllabus. In no case, however, may a student be withdrawn from a course whose absences are less than or equal to two times the number of weekly meeting days assigned to the course. A grade of an "F" will be recorded if a withdrawal occurs after the official withdrawal period.*

See "Class Attendance" at <https://coursecatalog.benedictine.edu/academic-regulations-procedures/>

In this course, the instructor may withdraw any student who misses more than 6 class meetings (and a grade of "F" will be recorded if the withdrawal occurs after the official withdrawal period). **In any case if the instructor chooses not to withdraw a student who misses more than 6 class meetings, the student will receive a final grade no higher than D.**

Grading: The graded coursework will consist of the following:

25%	100 points – 10 Homework Memos @ 10 points each
25%	100 points – 10 Tests @ 10 points each
25%	100 points – 1 Derivatives Project @ 100 points
25%	100 points – 1 Capital Budgeting Project @ 100 points
100%	400 points – Total

Final Grade: Final grades (out of a maximum of 400 points) will be determined as follows:

Grade	Percent		Points	
	Lower	Upper	Lower	Upper
A	92.50%	100.00%	370	400
A-	90.00%	92.49%	360	370
B+	87.50%	89.99%	350	360
B	82.50%	87.49%	330	350
B-	80.00%	82.49%	320	330
C+	77.50%	79.99%	310	320
C	72.50%	77.49%	290	310
C-	70.00%	72.49%	280	290
D	60.00%	69.99%	240	280
F	0.00%	60.00%	0	240

Schedule: The semester schedule of assigned readings, homework, case reports, and projects is provided at the end of this syllabus and on Canvas. This schedule may change as the semester progresses, and if that is the case, the instructor will provide students with an updated version.

Homework Memos: Students will complete 10 homework assignments. Instructions for each assignment are in the *Student Book of Handouts* and will also be available on Canvas, along with Excel templates and due dates. These dates are on the Semester Schedule. Each homework assignment is worth 10 points. Late assignments will be automatically penalized 3 points up to 24 hours past the deadline, and 5 points up to one week past the due date and at least 5 points thereafter. **No HW will be accepted when the deadline has passed by one week.**

Tests: Students will take 10 timed Tests on Canvas that consist primarily of multiple choice and true/false questions (from eBook chapters). Each test will be worth a total of 10 points. Due dates are listed on the Semester Schedule. **NO MAKE-UP TESTS WILL BE GIVEN if the student misses the deadline unless the instructor is notified in advance.**

Derivatives Project: A Derivatives Project (100 points) is due (see Semester Schedule) that requires graphing and analyzing currency Futures and Options contracts. Students will be given a memo describing this project and extensive time in class will be spent preparing for the required work.

Capital Budgeting Project: An International Capital Budgeting Project (100 points) is due at the end of the semester (See Semester Schedule). Students will be given a memo describing this project and extensive time in class will be spent preparing for the required work.

Grading Rubric on the 10 Homework Assignments (10 points each):

1. Overall quality of the submission = 5 points possible

Scale: Professional Grade with no blemishes = 5 points
Professional Grade with minor blemishes = 4 points
Decent but not Professional Grade = 3 points
Mediocre Effort = 2 points
Poor (i.e., sloppy) = 1 point
Obviously Lacking Care for Quality = 0 points

2. Completeness of the assignment and following directions = 3 points possible

Scale: Complete = 3 points
Incomplete = 1 to 2 points depending on extent of incompleteness
Obviously lacking effort = 0 points

3. Accuracy of the work produced = 2 points possible

Scale: Very effective = 2 points
Not very effective = 1 points
Obviously lacking effort = 0 points

Academic Misconduct: *Benedictine College endorses the principles of academic honesty. Any academic misconduct is contradictory to the purposes and welfare of both the student and the College. Academic misconduct includes, but is not limited to, cheating on examinations; plagiarizing; failing to properly document sources either intentionally or unintentionally; turning in work prepared by another source, falsifying data and documents; or gaining or giving unauthorized access to assessment materials. Using artificial intelligence (AI) for coursework in place of a student's own effort to learn constitutes academic misconduct. Therefore, students may use AI for their academics only when in accordance with instructor expectations and direction. Students should refer to the specified guidelines provided by their instructor.*

It is expected that the student will behave in a manner consistent with the values of Benedictine College in interactions in an academic context both inside and outside of the classroom. Should a student engage in inappropriate or aggressive language or behavior in relationship with faculty, or disruptive behavior in or out of the classroom, a faculty member may document the incident with the Vice Provost and the student may be subject to appropriate sanctions.

See <https://coursecatalog.benedictine.edu/academic-regulations-procedures/> for a description of Consequences and Due Process with respect to Academic Misconduct.

Reasonable Learning Accommodations: *Benedictine College complies with federal law requiring reasonable accommodations for individuals with documented disabilities that may affect their performance in a class or otherwise affect the individual's ability to enjoy the programs and services offered by the College. If a student requests an accommodation from a faculty member, the faculty member should direct the student to the Director of the Academic Research Center in the Student Success Center.*

See <https://coursecatalog.benedictine.edu/academic-services/> for information on Student Academic Services and Learning Accommodations.

Teaching Quality: Your professor pledges to

- Follow the course schedule.
- Start and end every class on time.
- Keep grades on Canvas updated for each student's progress.
- Be available to students outside of class times.
- Teach with enthusiasm, respect, and confidence in the students.

Classroom Decorum: In order to create and maintain a classroom environment that promotes the highest quality of teaching and learning, students are asked to adhere to the following rules:

- Dress in a manner that reflects a serious work environment
- DO NOT bring meals into the classroom (snacks, bottled drinks and coffee are allowed)
- DO NOT to leave the classroom (and return) unless it is an emergency
- DO NOT use computers for anything but class work
- Put away cell phones and personal devices AT ALL TIMES

FINC-4900: International Finance - Semester Schedule - Fall 2026

as of 8/26/2026

Week	Class	Day	Date	Topics and lecture topics	HW # Due	Test # Due	Project # Due
1	1	Wed	26-Aug	Review Syllabus and Semester Schedule			
	2	Fri	28-Aug	Introduction: International Monetary Order (pp. 1-14)			
2	3	Mon	31-Aug	Topic 1: Globalization and the Multnational Firm	N/A	N/A	
	4	Wed	2-Sep	Review HW #1 and Test #1 (Chapter 1 in eBook)			
	5	Fri	4-Sep	Review Class Only - Time allotted to take Test 1 on Canvas			
3	6	Mon	7-Sep	Topic 2: International Flow of Funds and Exchange Rates	1	1	
	7	Wed	9-Sep	Review HW #2 and Test #2 (Chapter 2 in eBook)			
	8	Fri	11-Sep	Review Class Only - Time allotted to take Test 2 on Canvas			
4	9	Mon	14-Sep	Topic 3: Foreign Exchange Markets	2	2	
	10	Wed	16-Sep	Review HW #3 and Test #3 (Chapter 3 in eBook)			
	11	Fri	18-Sep	Review Class Only - Time allotted to take Test 3 on Canvas			
5	12	Mon	21-Sep	Topic 4: Exchange Rate Determination	3	3	
	13	Wed	23-Sep	Review HW #4 and Test #4 (Chapter 4 in eBook)			
	14	Fri	25-Sep	Review Class Only - Time allotted to take Test 4 on Canvas			
6	15	Mon	28-Sep	Topic 5: Currency Derivatives: Forwards, Futures, Options	4	4	
	16	Wed	30-Sep	Review HW #5 and Test #5 (Chapter 5 in eBook)			
	17	Fri	2-Oct	Review Class Only - Time allotted to take Test 5 on Canvas			
7	18	Mon	5-Oct	Project 1: Currency Derivatives: Forwards, Futures, Options Handouts (No Test)	5	5	
	19	Wed	7-Oct				
	20	Fri	9-Oct				
8	21	Mon	12-Oct	Project 1: Currency Derivatives (continued)	N/A	N/A	
	22	Wed	14-Oct				1
			Fri	16-Oct			Fall Break - No Classes
9	23	Mon	19-Oct	Topic 6: Government Influence on Exchange Rates	N/A	N/A	
	24	Wed	21-Oct	Review HW #6 and Test #6 (Chapter 6 in eBook)			
	25	Fri	23-Oct	Review Class Only - Time allotted to take Test 6 on Canvas			
10	26	Mon	26-Oct	Topic 7: Int'l Arbitrage: Locational, Triangular, Covered-Interest	6	6	
	27	Wed	28-Oct	Review HW #7 and Test #7 (Chapter 7 in eBook)			
	28	Fri	30-Oct	Review Class Only - Time allotted to take Test 7 on Canvas			
11	29	Mon	2-Nov	Topic 8: Int'l Parity: Inflation, Interest Rates, Exchange Rates	7	7	
	30	Wed	4-Nov	Review HW #8 and Test #8 (Chapter 8 in eBook)			
	31	Fri	6-Nov	Review Class Only - Time allotted to take Test 8 on Canvas			
12	32	Mon	9-Nov	Topic 9: Managing Transaction Exposure with Money Markets	8	8	
	33	Wed	11-Nov	Review HW #9 and Test #9 (Chapter 11 in eBook)			
	34	Fri	13-Nov	Review Class Only - Time allotted to take Test 9 on Canvas			
13	35	Mon	16-Nov	Topic 10: Managing Economic and Translation Risk Exposure	9	9	
	36	Wed	18-Nov	Review HW #10 and Test #10 (Chapter 12 in eBook)			
	37	Fri	20-Nov	Review Class Only - Time allotted to take Test 10 on Canvas			
14	38	Mon	23-Nov	Topic 11: International Capital Budgeting	10	10	
		Wed	25-Nov	Thanksgiving Break - No Classes			
		Fri	27-Nov				
15	39	Mon	30-Nov	Project 2: International Capital Budgeting Handouts (No Test)	N/A	N/A	
	40	Wed	2-Dec				
	41	Fri	4-Dec				
16	42	Mon	7-Dec	Review Project #2	N/A	N/A	
		Wed	9-Dec	Study Day - No Class			
		Thur	10-Dec	Final Exam Time Slot: 8:00 to 9:50 AM			2